

JOSE RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES AND ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CPE C313

ARTIFICIAL INTELLIGENCE

LABORATORY

1&2

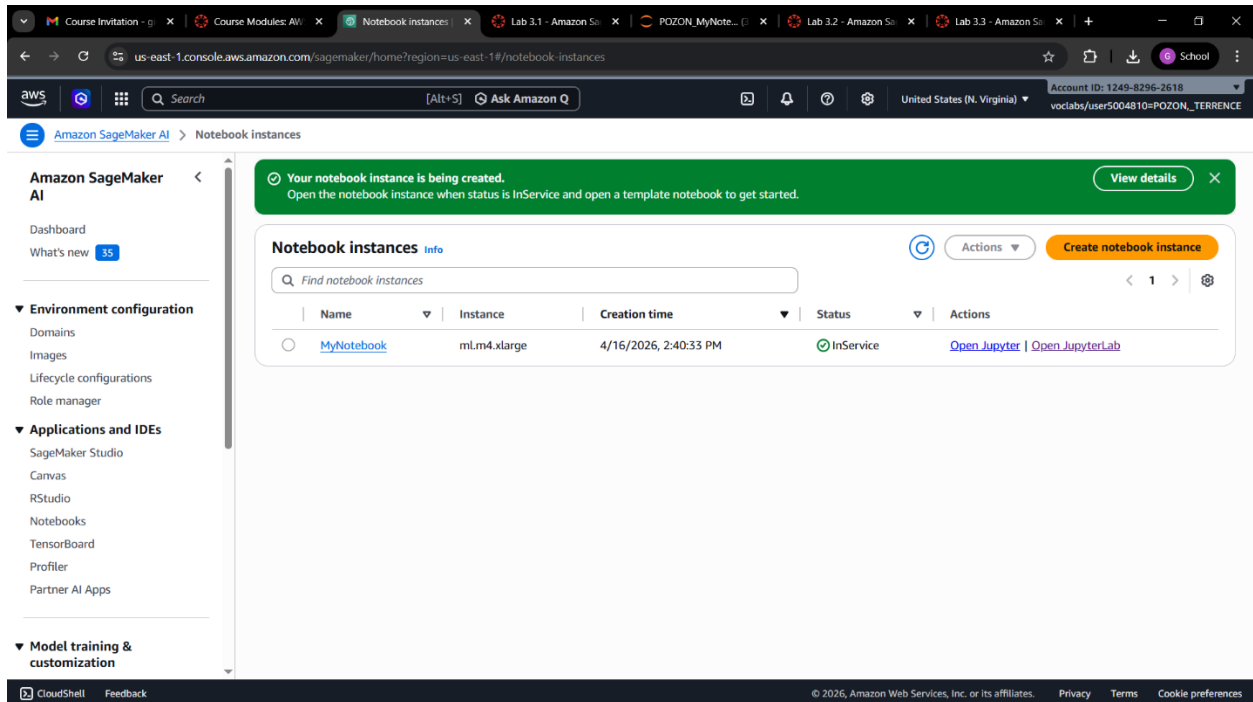
Pozon, Gion Terrence B.

301G

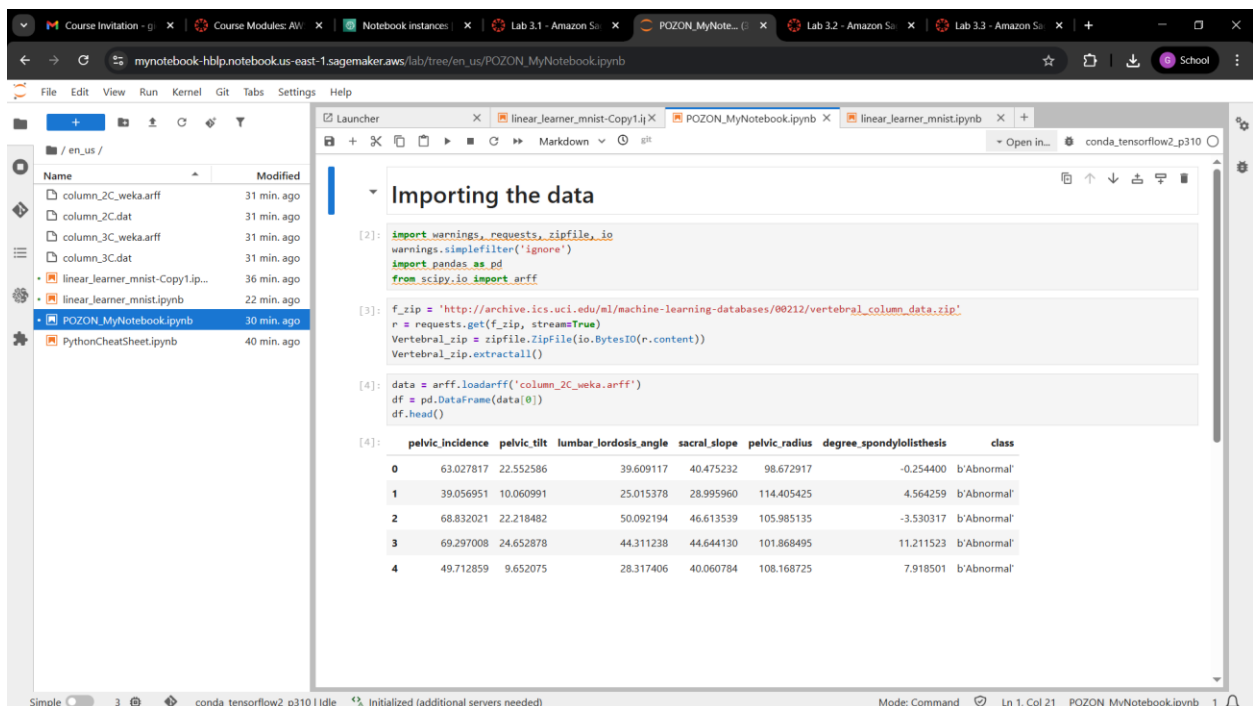
2026-04-16

LABORATORY 1

1. AWS Console account
2. MyNotebook instance configuration



3. Output after the code below is executed



4. Screenshot of the Submitted activity.

The screenshot displays the AWS Academy Instructure LMS interface. The browser address bar shows the URL `awsacademy.instructure.com/courses/168322/modules/items/16494070`. The page title is "AMLFv1EN-LT13-168322 > Modules > Module 3 - Implementing a Machine Learning pipeline with Amazon SageMaker > Lab 3.1 - Amazon SageMaker - Creating and importing data".

On the left is a dark sidebar with navigation links: Home, Modules, Discussions, Grades, Lucid (Whiteboard), Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area has a top bar with a "Submit" button and tabs for "Details", "AWS", "Start Lab", "End Lab", "0:55", "Grades", and "Actions". Below this is a toolbar with "Files", "README", "Terminal", and "Source" options.

The main content area contains a text editor with a dropdown menu set to "EN-US". The text in the editor includes instructions: "On this page, choose `End Lab`, and then choose `Yes`." and "A panel should appear with this message: *DELETE has been initiated... You may close this message box now.*". Below this, it says "41. To close the panel, choose the **X** in the top-right corner." and a copyright notice: "©2022 Amazon Web Services, Inc. and its affiliates. All rights reserved. This work may not be reproduced or redistributed, in whole or in part, without prior written permission from Amazon Web Services, Inc. Commercial copying, lending, or selling is prohibited."

On the right side of the main content area is a large gray box with a blue "Launch Terminal" button. At the bottom right of this box is a small gray notification box that says "Submission recorded" with a close button (X).

At the bottom of the page are "Previous" and "Next" navigation buttons.

LABORATORY 2

1. Output of this last task:

- Examined the structure of a dataset by using pandas

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The notebook is titled "Step 1: Exploring the data". The code in the cells is as follows:

```
[4]: df.shape
[4]: (310, 7)

You will now get a list of the columns.

[5]: df.columns
[5]: Index(['pelvic_incidence', 'pelvic_tilt', 'lumbar_lordosis_angle',
'sacral_slope', 'pelvic_radius', 'degree_spondylolisthesis', 'class'],
dtype='object')

You can see the six biomechanical features, and the target column is named class.

What column types do you have?

[6]: df.dtypes
[6]: pelvic_incidence    float64
pelvic_tilt             float64
lumbar_lordosis_angle   float64
sacral_slope            float64
pelvic_radius           float64
degree_spondylolisthesis float64
class                  object
dtype: object
```

- Viewed statistics with pandas and Matplotlib

The screenshot shows the same Jupyter Notebook interface, but now with a line plot. The code in the cells is as follows:

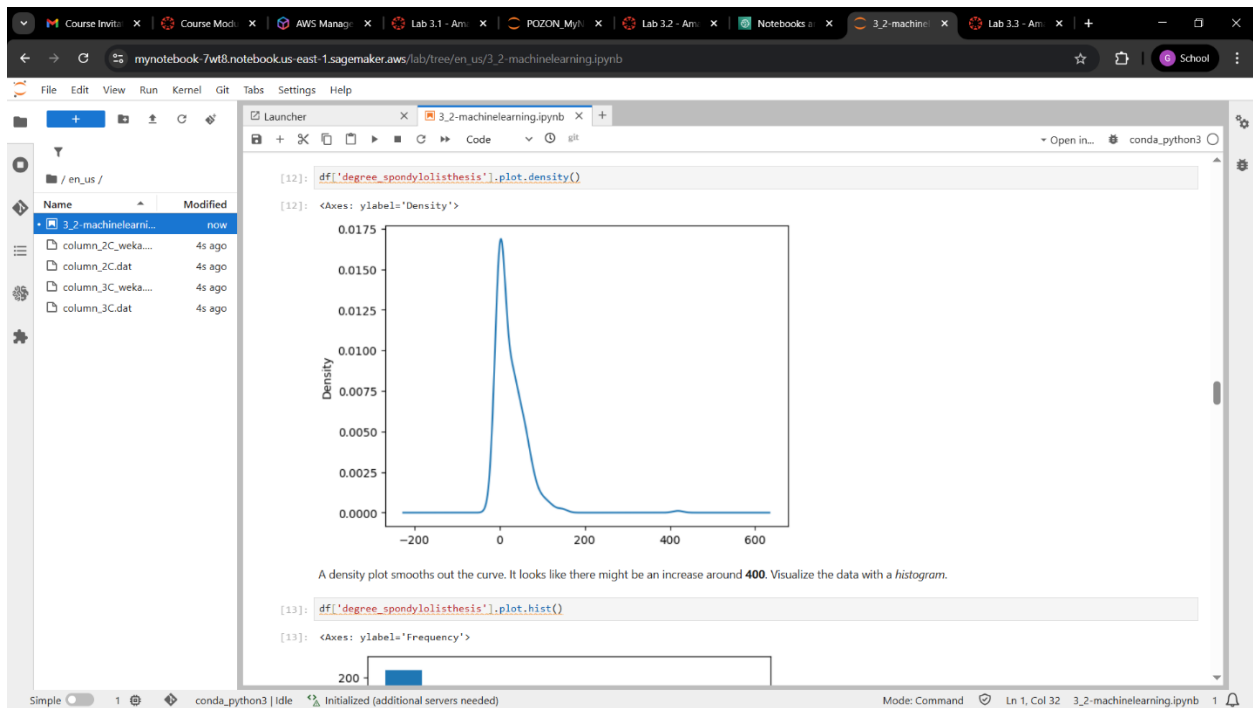
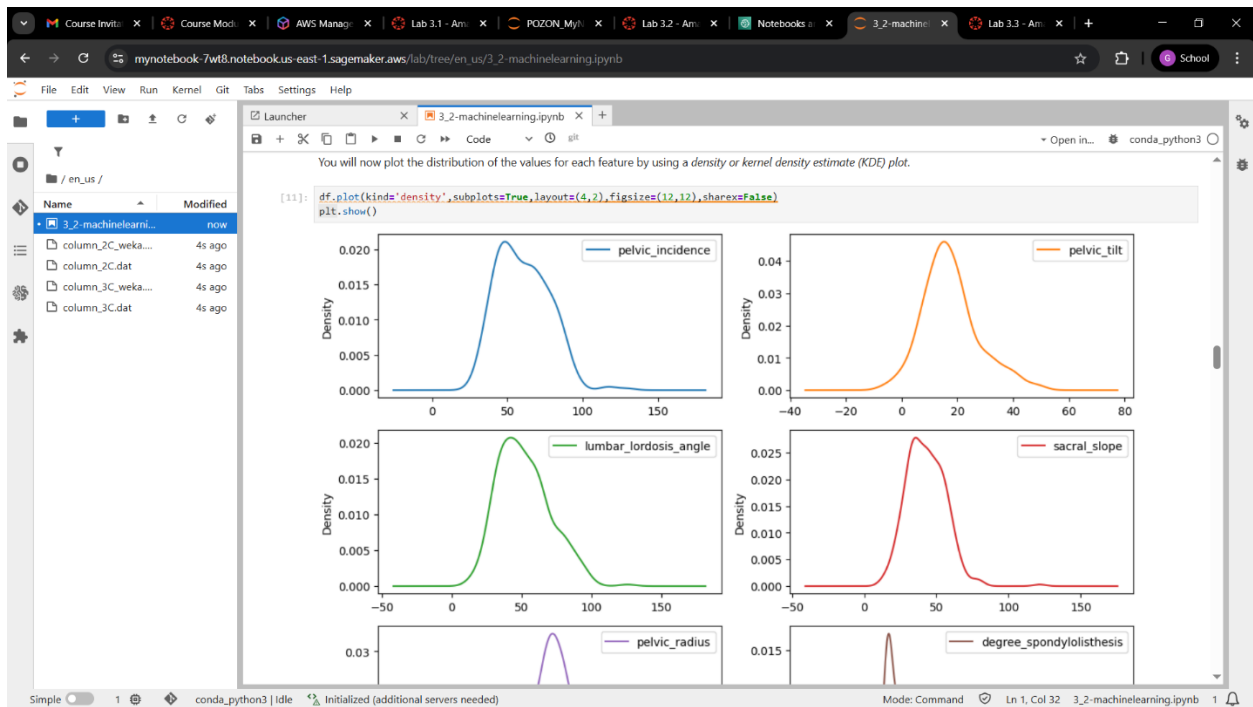
```
[10]: import matplotlib.pyplot as plt
      %matplotlib inline
      df.plot()

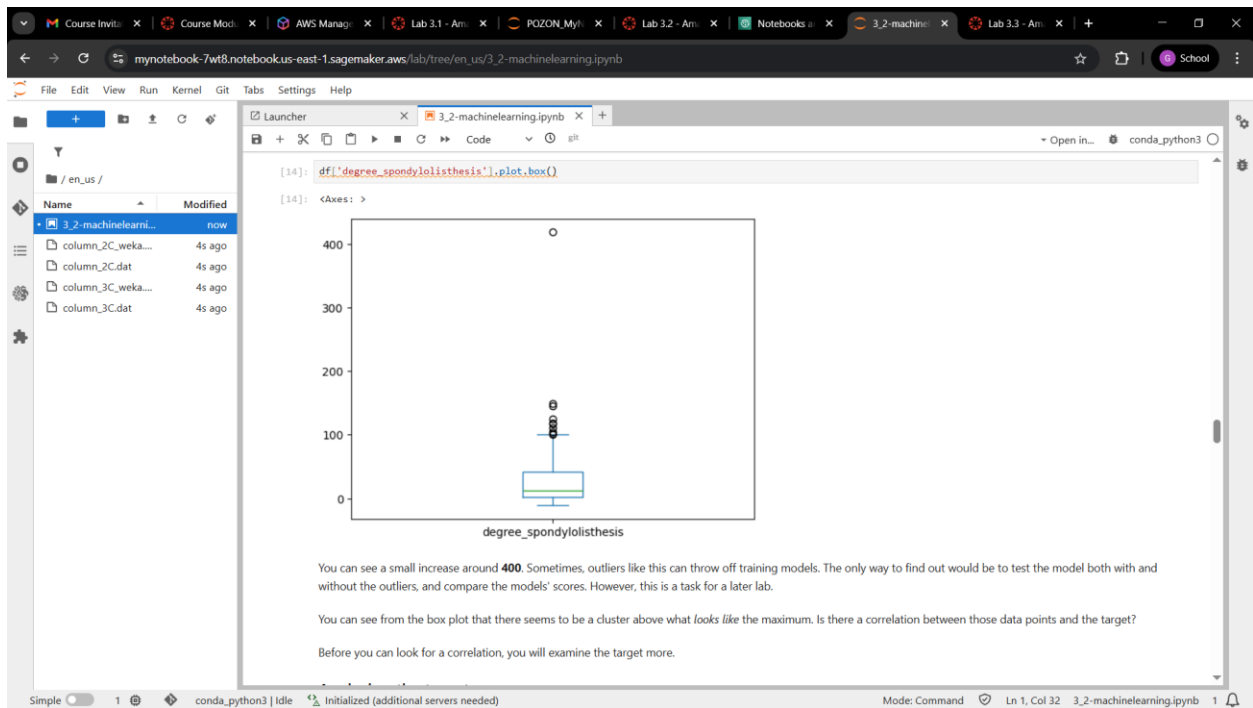
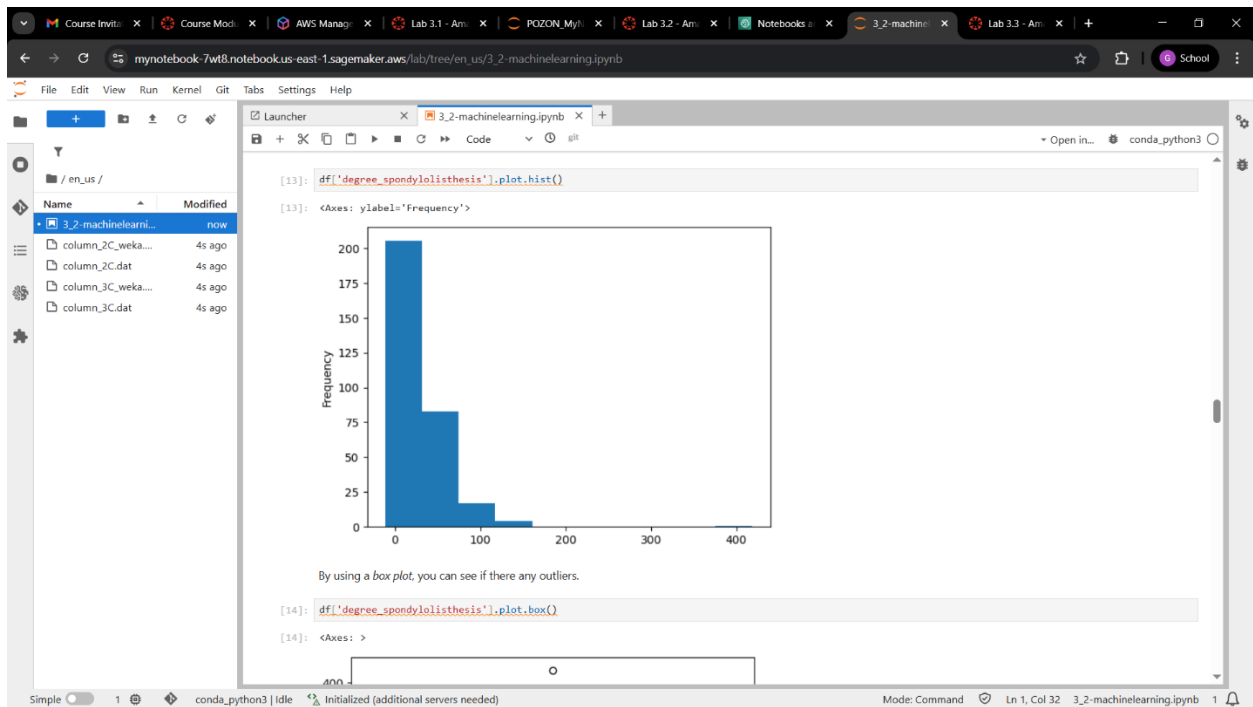
[10]: <Axes: >
```

The plot displays six features over an index range of 0 to 300. The y-axis ranges from 0 to 400. The features are: pelvic_incidence (blue), pelvic_tilt (orange), lumbar_lordosis_angle (green), sacral_slope (red), pelvic_radius (purple), and degree_spondylolisthesis (brown). The plot shows that most features have values between 0 and 100, while pelvic_radius has a much higher range, peaking near 400.

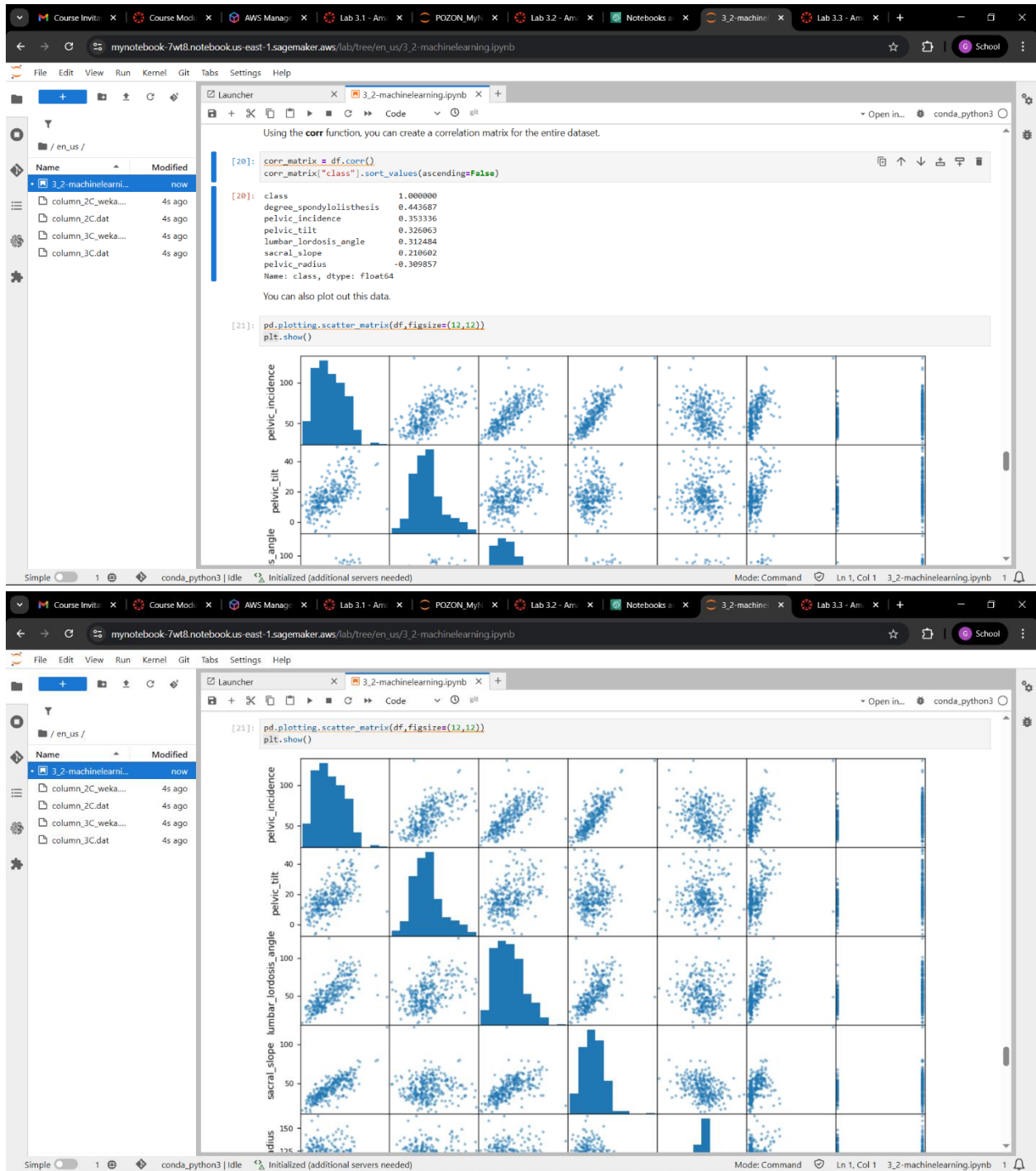
You will now plot the distribution of the values for each feature by using a density or kernel density estimate (KDE) plot.

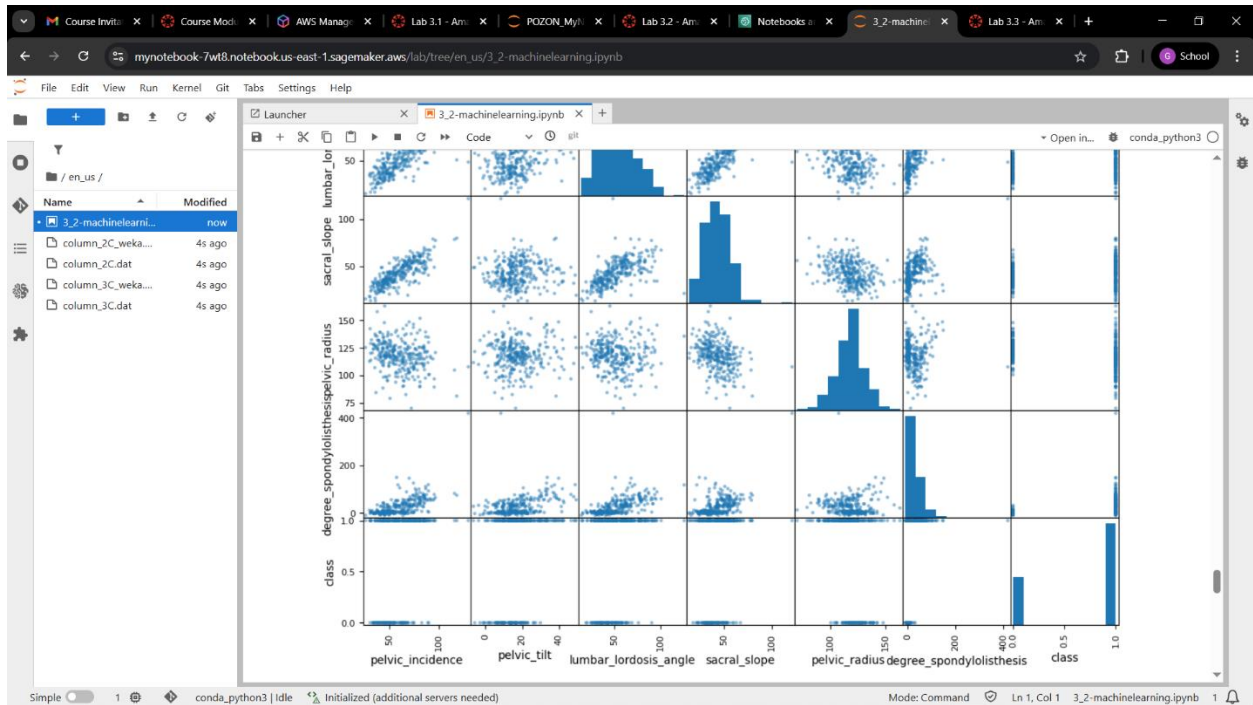
```
[11]: df.plot(kind='density', subplot=True, layout=(4,2), figsize=(12,12), sharex=False)
      plt.show()
```



- Looked for correlation between features in a dataset





2. Explain the correlation of the different variables:

- Question:** Are there any features that aren't well-distributed? Are there any features with outliers that you want to look at? Does it look like there might be any correlations between features?
 - `degree_spondylolisthesis` has the most outliers since its max is almost 10 times the value of 75% percentile, meaning there are 25% of people that have the values between 41 and 418, `lumbar_lordosis_angle`, `pelvic_tilt`, `pelvic_incidence` have similar values with each other which can indicate correlation
- Which variables are highly correlated?
 - `lumbar_lordosis_angle`, `pelvic_tilt`, `pelvic_incidence` since they are closely together with the values of 0.31, 0.32, and 0.35 respectively
- Which variables may have little or no effect on the target variables.
 - `pelvic_radius` since it has negative value which is -0.30

Canvas Details

The screenshot displays the AWS Academy web application. The browser's address bar shows the URL `awsacademy.instructure.com/courses/168322/modules/items/16494074`. The page title is "AMLFv1EN-LT13-168322 > Modules > Module 3 - Implementing a Machine Learning pipeline with Amazon SageMaker > Lab 3.2 - Amazon SageMaker - Exploring Data".

On the left sidebar, there are navigation links: Home, Modules, Discussions, Grades (with a notification badge), and Lucid (Whiteboard). Below these are icons for Account, Dashboard, Courses, Calendar, Inbox, History, and Help.

The main content area shows a lab details page. A "Credentials" modal is open, displaying the following information:

- Cloud Access**
- AWS CLI:** [Show]
- Cloud Labs**
 - Remaining session time: 01:54:11 (115 minutes)
 - Session started at: 2026-04-16T03:16:34-0700
 - Session to end at: 2026-04-16T05:16:34-0700
- Accumulated lab time: 02:05:00 (125 minutes)
- No running instance
- SSH key:** [Show] [Download PEM] [Download PPK]
- AWS SSO:** [Download URL]

At the bottom of the page, there are "Previous" and "Next" navigation buttons.

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CPE C313

ARTIFICIAL INTELLIGENCE

LABORATORY

3 & 4

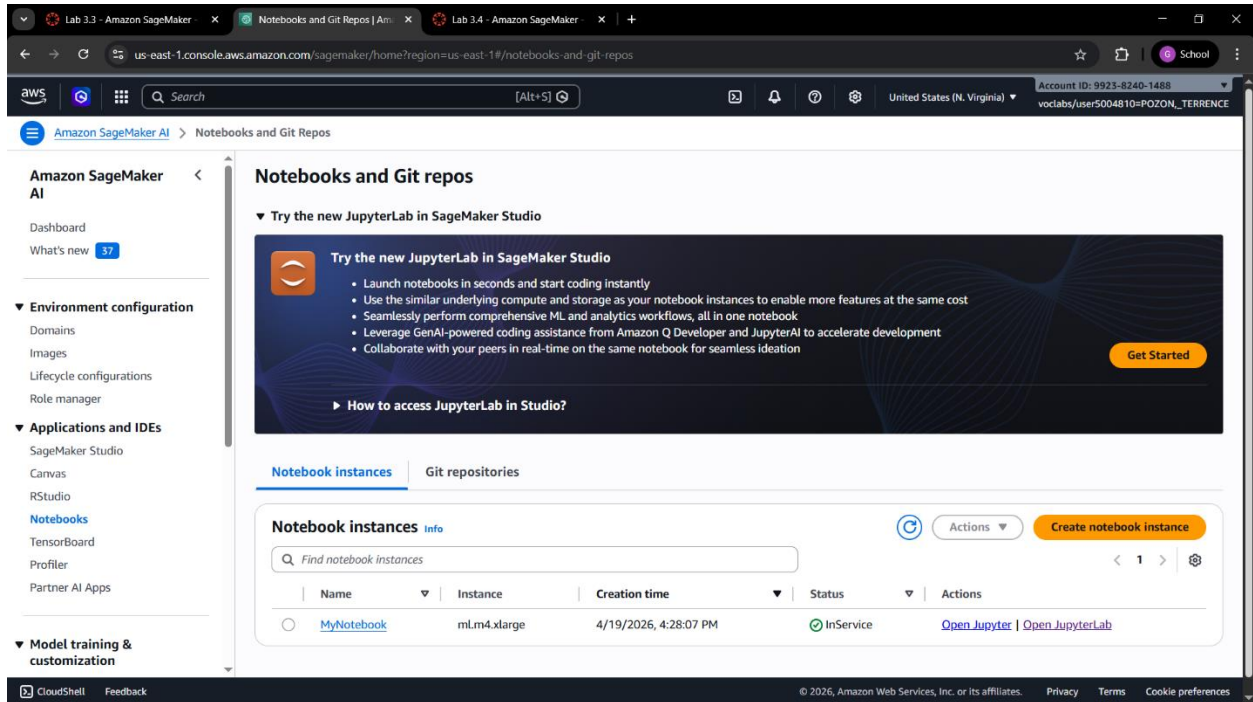
Pozon, Gion Terrence B.

301G

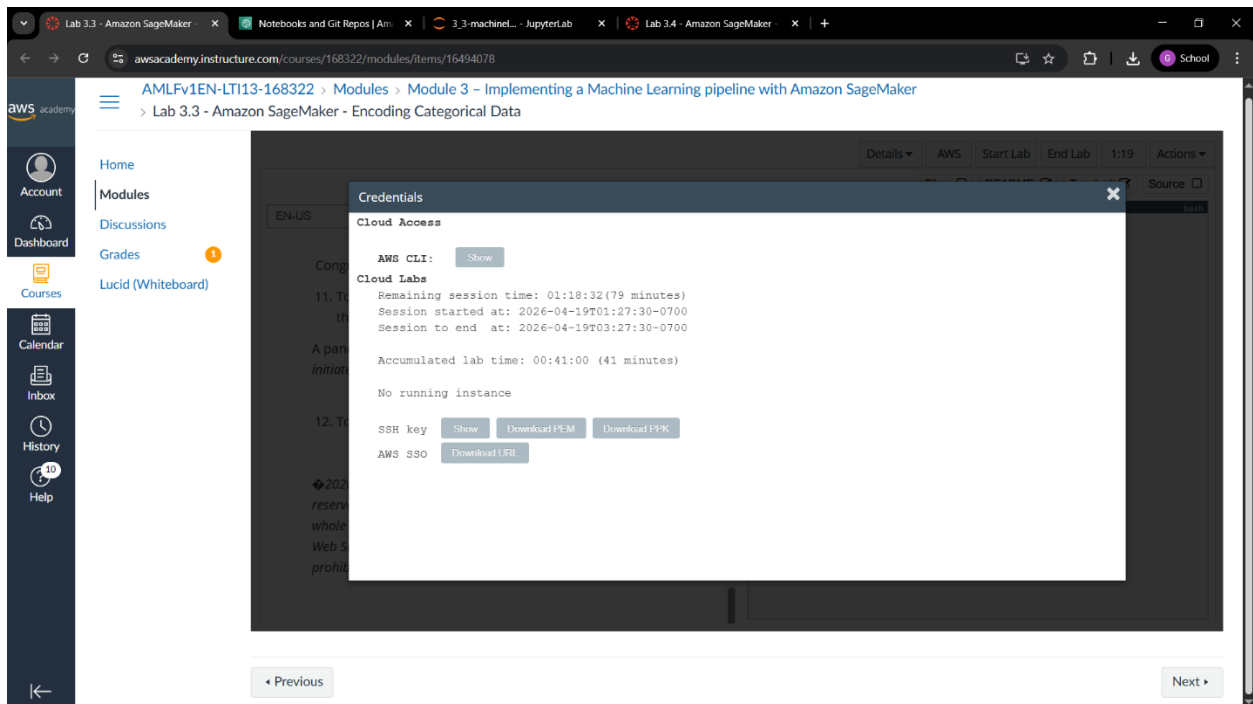
2026-04-19

Laboratory Activity 3

AWS Console account & MyNotebook instance configuration



Canvas Screenshots



Challenge Task

Lab 3.3 - Amazon SageMaker | x | Notebooks and Git Repos | Am | x | 3.3-machinelearning.ipynb | x | Lab 3.4 - Amazon SageMaker | x | +

mynotebook-c3s2.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3.3-machinelearning.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Launcher x 3.3-machinelearning.ipynb x +

Open in... conda_python3

Challenge task: Go back to the beginning of this lab, and add other columns to the dataset. How would you encode the values of each column? Update the code to include some of the other features.

```
[27]: df_car2 = pd.read_csv(url1, sep=';', names=col_names, na_values="?", headers=None)

df_car2 = df_car2[['aspiration', 'num-of-doors', 'drive-wheels', 'num-of-cylinders',
                  'body-style', 'engine-type', 'fuel-type', 'engine-location',
                  'horsepower', 'price']].copy()

# Ordinal
df_car2['doors'] = df_car2['num-of-doors'].replace(door_mapper)
df_car2['cylinders'] = df_car2['num-of-cylinders'].replace(cylinder_mapper)

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns='aspiration', drop_first=True)

# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns='drive-wheels')

# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns='body-style')
df_car2 = pd.get_dummies(df_car2, columns='engine-type')

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns='fuel-type', drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns='engine-location', drop_first=True)

# Numeric: horsepower, price

df_car2.head()

/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'
df_car2['doors'] = df_car2['num-of-doors'].replace(door_mapper)

/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'
df_car2['cylinders'] = df_car2['num-of-cylinders'].replace(cylinder_mapper)
```

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 21, Col 26 3.3-machinelearning.ipynb 1

Lab 3.3 - Amazon SageMaker | x | Notebooks and Git Repos | Am | x | 3.3-machinelearning.ipynb | x | Lab 3.4 - Amazon SageMaker | x | +

mynotebook-c3s2.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3.3-machinelearning.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Launcher x 3.3-machinelearning.ipynb x +

Open in... conda_python3

```
# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns='body-style')
df_car2 = pd.get_dummies(df_car2, columns='engine-type')

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns='fuel-type', drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns='engine-location', drop_first=True)

# Numeric: horsepower, price

df_car2.head()

/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'
df_car2['doors'] = df_car2['num-of-doors'].replace(door_mapper)

/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'
df_car2['cylinders'] = df_car2['num-of-cylinders'].replace(cylinder_mapper)
```

```
[27]:
```

	num-of-doors	num-of-cylinders	horsepower	price	doors	cylinders	aspiration	turbo	drive-wheels_4wd	drive-wheels_fwd	drive-wheels_rwd	body-style_convertible	body-style_hardtop	body-style_hatchback
0	two	four	111.0	13495.0	2.0	4	False	False	False	True	True	False	False	False
1	two	four	111.0	16500.0	2.0	4	False	False	False	True	True	False	False	False
2	two	six	154.0	16500.0	2.0	6	False	False	False	True	True	False	False	True
3	four	four	102.0	13950.0	4.0	4	False	False	True	False	False	False	False	False
4	four	five	115.0	17450.0	4.0	5	False	True	False	False	False	False	False	False

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 21, Col 26 3.3-machinelearning.ipynb 1

Lab 3.3 - Amazon SageMaker x Notebooks and Git Repos | Am x 3.3-machinelearning.ipynb x JupyterLab x Lab 3.4 - Amazon SageMaker x +

mynotebook-c3s2.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3.3-machinelearning.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Launcher x 3.3-machinelearning.ipynb x +

conda_python3

```
# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns=['body-style'])
df_car2 = pd.get_dummies(df_car2, columns=['engine-type'])

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns=['fuel-type'], drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns=['engine-location'], drop_first=True)

# Numeric: horsepower, price

df_car2.head()
```

/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'

/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'

[27]:

drive-wheels_4wd	drive-wheels_fwd	drive-wheels_rwd	body-style_convertible	body-style_hardtop	body-style_hatchback	body-style_sedan	body-style_wagon	engine-type_dohc	engine-type_dohcv	engine-type_i	engine-type_ohc	engine-type_ohv
False	False	True	True	False	False	False	False	True	False	False	False	Fals
False	False	True	True	False	False	False	False	True	False	False	False	Fals
False	False	True	False	False	True	False	False	False	False	False	False	Fals
False	True	False	False	False	False	True	False	False	False	False	True	Fals
True	False	False	False	False	False	True	False	False	False	False	True	Fals

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 21, Col 26 3.3-machinelearning.ipynb 1

Lab 3.3 - Amazon SageMaker x Notebooks and Git Repos | Am x 3.3-machinelearning.ipynb x JupyterLab x Lab 3.4 - Amazon SageMaker x +

mynotebook-c3s2.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/3.3-machinelearning.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Launcher x 3.3-machinelearning.ipynb x +

conda_python3

```
# Non-ordinal, multiple values
df_car2 = pd.get_dummies(df_car2, columns=['body-style'])
df_car2 = pd.get_dummies(df_car2, columns=['engine-type'])

# Non-ordinal, two values
df_car2 = pd.get_dummies(df_car2, columns=['fuel-type'], drop_first=True)
df_car2 = pd.get_dummies(df_car2, columns=['engine-location'], drop_first=True)

# Numeric: horsepower, price

df_car2.head()
```

/tmp/ipykernel_13520/2816081130.py:8: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'

/tmp/ipykernel_13520/2816081130.py:9: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.infer_objects(copy=False)'. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcasting', True)'

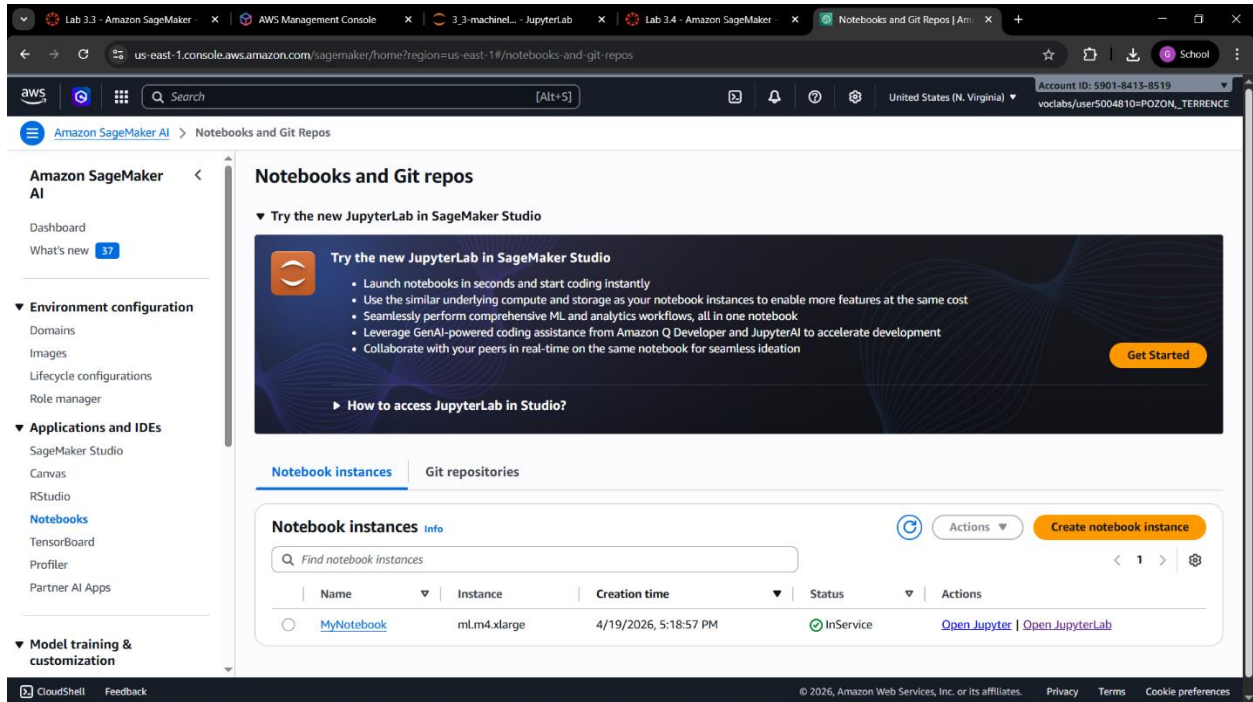
[27]:

body-style_convertible	body-style_hardtop	body-style_hatchback	body-style_sedan	body-style_wagon	engine-type_dohc	engine-type_dohcv	engine-type_i	engine-type_ohc	engine-type_ohcv	engine-type_rotor	fuel-type_gas	engine-location_rear
True	False	False	False	False	True	False	False	False	False	False	True	False
True	False	False	False	False	True	False	False	False	False	False	True	False
False	False	True	False	False	False	True	False	False	True	False	True	False
False	False	False	True	False	False	False	True	False	False	False	True	False
False	False	False	True	False	False	False	True	False	False	False	True	False

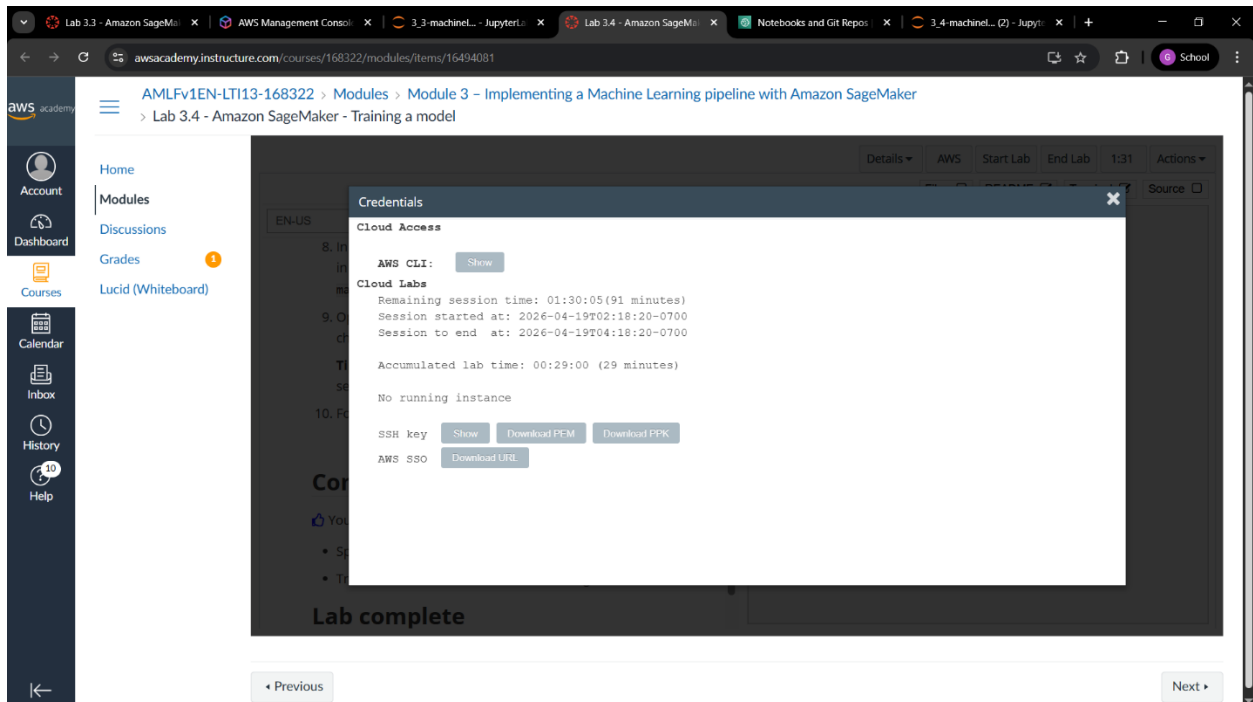
Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 21, Col 26 3.3-machinelearning.ipynb 1

Laboratory Activity 4

AWS Console account & MyNotebook instance configuration



Canvas Screenshot



Model Trained

The screenshot displays the Amazon SageMaker JupyterLab interface. On the left, a file browser shows the directory structure with files like `column_2C.weka...`, `column_2C.dat`, `column_3C.weka...`, `column_3C.dat`, `linear_learner_mni...`, and `PythonCheatSheet...`. The main area shows a notebook with the following content:

```
"s3://{1}/{1}/train/".format(bucket,prefix,train_file),
content_type='text/csv')

validate_channel = sagemaker.inputs.TrainingInput(
    "s3://{1}/{1}/validate/".format(bucket,prefix,validate_file),
    content_type='text/csv')

data_channels = {'train': train_channel, 'validation': validate_channel}
```

Running `fit` will train the model.

Note: This process can take up to 5 minutes.

```
[19]: xgb_model.fit(inputsdata_channels, logs=False)
```

INFO:sagemaker:Creating training-job with name: sagemaker-xgboost-2026-04-19-09-42-30-044

2026-04-19 09:42:31 Starting - Starting the training job..
2026-04-19 09:42:45 Starting - Preparing the instances for training...
2026-04-19 09:43:08 Downloading - Downloading input data.....
2026-04-19 09:44:09 Downloading - Downloading the training image...
2026-04-19 09:44:29 Training - Training image download completed. Training in progress....
2026-04-19 09:44:50 Uploading - Uploading generated training model...
2026-04-19 09:45:08 Completed - Training job completed

After the training is complete, you are ready to test and evaluate the model. However, you will do testing and validation in later labs.

Congratulations!

You have completed this lab, and you can now end the lab by following the lab guide instructions.

The bottom status bar indicates the environment is `conda_python3` and the kernel is `Idle`. The mode is `Edit`, and the cursor is at `Ln 1, Col 1` in the file `3_4-machinelearning.ipynb`.

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CPE C313

ARTIFICIAL INTELLIGENCE

LABORATORY

3.5 - 3.7

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2026-04-29

Activity Lab 3.5

AWS Console account & MyNotebook

Amazon SageMaker AI

Dashboard

What's new **42**

▼ Environment configuration

- Domains
- Images
- Lifecycle configurations
- Role manager

▼ Applications and IDEs

- SageMaker Studio
- Canvas
- RStudio
- Notebooks**
- TensorBoard
- Profiler
- Partner AI Apps

▼ Model training & customization

Notebooks and Git repos

▼ Try the new JupyterLab in SageMaker Studio

Try the new JupyterLab in SageMaker Studio

- Launch notebooks in seconds and start coding instantly
- Use the similar underlying compute and storage as your notebook instances to enable more features at the same cost
- Seamlessly perform comprehensive ML and analytics workflows, all in one notebook
- Leverage GenAI-powered coding assistance from Amazon Q Developer and JupyterAI to accelerate development
- Collaborate with your peers in real-time on the same notebook for seamless ideation

▶ How to access JupyterLab in Studio?

Notebook instances **Git repositories**

Notebook instances Info

Find notebook instances

Actions **Create notebook instance**

Name	Instance	Creation time	Status	Actions
MyNotebook	ml.m4.xlarge	4/29/2026, 12:04:41 AM	InService	Open Jupyter Open JupyterLab

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Challenge Task 1

File Edit View Run Kernel Git Tabs Settings Help

Launcher

3_5-machinelearning.ipynb

Open in... conda_python3

136	1	88.024499	39.844669	81.774473	48.179830	116.601538	56.766083
230	0	65.611802	23.137919	62.582179	42.473883	124.128001	-4.083298
134	1	52.204693	17.212673	78.094969	34.992020	136.972517	54.939134
130	1	50.066786	9.120340	32.168463	40.946446	99.712453	26.766697
47	1	41.352504	16.577364	30.706191	24.775141	113.266675	-4.497958

Question: Is the prediction accurate?

- Yes, because it got 4 out of 5 correct with an accuracy of 80%

Challenge task: Update the previous code to send the second row of the dataset. Are those predictions correct? Try this task with a few other rows.

It can be tedious to send these rows one at a time. You could write a function to submit these values in a batch, but SageMaker already has a batch capability. You will examine that feature next. However, before you do, you will terminate the model.

```
[11]: #Challenge_Task_1
for i in range(5):
    buf = io.StringIO()
    test.iloc[i:i+1, 1:].to_csv(buf, header=False, index=False)
    pred = xgb_predictor.predict(buf.getvalue())
    actual = test.iloc[i]['class']
    pred_binary = 1 if float(pred) > 0.65 else 0
    print(f"Row {i} | Predicted: {pred_binary} (score: {float(pred):.4f}) | Actual: {int(actual)} | {'correct' if pred_binary == actual else 'incorrect'}")

Row 0 | Predicted: 1 (score: 0.9966) | Actual: 1 | correct
Row 1 | Predicted: 1 (score: 0.7773) | Actual: 0 | incorrect
Row 2 | Predicted: 1 (score: 0.9946) | Actual: 1 | correct
Row 3 | Predicted: 1 (score: 0.9937) | Actual: 1 | correct
Row 4 | Predicted: 1 (score: 0.9391) | Actual: 1 | correct
```

Step 3: Terminating the deployed model

Simple 1 conda_python3 | Busy Fully initialized Mode: Command Ln 2, Col 66 3_5-machinelearning.ipynb 1

Challenge Task 2

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The file explorer shows a directory structure with files like 'column_2C_weka...', 'column_3C_weka...', 'column_3C.dat', 'linear_learner_mni...', and 'PythonCheatSheet...'. The code editor displays a table of data with columns: 'id', 'class', 'v1', 'v2', 'v3', 'v4', 'v5', 'v6', 'v7', 'v8', 'v9', 'v10', 'v11', 'v12', 'v13', 'v14', 'v15', 'v16', 'v17', 'v18', 'v19', 'v20', 'v21', 'v22', 'v23', 'v24', 'v25', 'v26', 'v27', 'v28', 'v29', 'v30', 'v31', 'v32', 'v33', 'v34', 'v35', 'v36', 'v37', 'v38', 'v39', 'v40', 'v41', 'v42', 'v43', 'v44', 'v45', 'v46', 'v47', 'v48', 'v49', 'v50', 'v51', 'v52', 'v53', 'v54', 'v55', 'v56', 'v57', 'v58', 'v59', 'v60', 'v61', 'v62', 'v63', 'v64', 'v65', 'v66', 'v67', 'v68', 'v69', 'v70', 'v71', 'v72', 'v73', 'v74', 'v75', 'v76', 'v77', 'v78', 'v79', 'v80', 'v81', 'v82', 'v83', 'v84', 'v85', 'v86', 'v87', 'v88', 'v89', 'v90', 'v91', 'v92', 'v93', 'v94', 'v95', 'v96', 'v97', 'v98', 'v99', 'v100'. The table contains 100 rows of data. Below the table, there is a challenge task section with a note and a code cell. The note states: 'Note: The threshold in the binary_convert function is set to .65. Challenge task: Experiment with changing the value of the threshold. Does it impact the results?'. The code cell contains the following code:

```
[10]: # Challenge Task 2
for threshold in [0.3, 0.5, 0.65, 0.8]:
    def binary_convert(x, tthreshold):
        return 1 if x > t else 0

    target_predicted['binary'] = target_predicted['class'].apply(binary_convert)
    correct = (target_predicted['binary'].values == test['class'].values).sum()
    print(f'Threshold {threshold:.2f} | Accuracy: {correct}/{len(test)} = {correct/len(test):.2%}')
```

Threshold 0.30 | Accuracy: 26/31 = 83.87%
Threshold 0.50 | Accuracy: 26/31 = 83.87%
Threshold 0.65 | Accuracy: 25/31 = 80.65%
Threshold 0.80 | Accuracy: 25/31 = 80.65%

Congratulations!

You have completed this lab, and you can now end the lab by following the lab guide instructions.

Last Codes

The screenshot shows the same Jupyter Notebook interface as the previous one, but with the code cell expanded to show the full code. The code is the same as in the previous screenshot, but the output is now visible. The output shows the accuracy for each threshold value. The code is as follows:

```
[10]: # Challenge Task 2
for threshold in [0.3, 0.5, 0.65, 0.8]:
    def binary_convert(x, tthreshold):
        return 1 if x > t else 0

    target_predicted['binary'] = target_predicted['class'].apply(binary_convert)
    correct = (target_predicted['binary'].values == test['class'].values).sum()
    print(f'Threshold {threshold:.2f} | Accuracy: {correct}/{len(test)} = {correct/len(test):.2%}')
```

Threshold 0.30 | Accuracy: 26/31 = 83.87%
Threshold 0.50 | Accuracy: 26/31 = 83.87%
Threshold 0.65 | Accuracy: 25/31 = 80.65%
Threshold 0.80 | Accuracy: 25/31 = 80.65%

Congratulations!

You have completed this lab, and you can now end the lab by following the lab guide instructions.

Canvas Details

The screenshot displays the AWS Academy Canvas interface. The browser address bar shows the URL `awsacademy.instructure.com/courses/168322/modules/items/16494083`. The page title is **AMLFv1EN-LTI13-168322 > Modules > Module 3 - Implementing a Machine Learning pipeline with Amazon SageMaker**, with a sub-header **> Lab 3.5 - Amazon SageMaker - Deploying a model**.

The left sidebar contains navigation links: Home, Modules, Discussions, Grades (with a notification badge), and Lucid (Whiteboard). The main content area shows a 'Credentials' modal window with the following details:

- Cloud Access**: Includes an 'AWS CLI' section with a 'Show' button.
- Cloud Labs**: Displays session information: 'Remaining session time: 00:36:37 (37 minutes)', 'Session started at: 2026-04-28T09:04:05-0700', and 'Session to end at: 2026-04-28T11:04:05-0700'. It also shows 'Accumulated lab time: 01:23:00 (83 minutes)' and 'No running instance'.
- SSH key**: Includes a 'Show' button and 'Download PEM' and 'Download PPK' buttons.
- AWS SSO**: Includes a 'Download UGI' button.

The modal window has a close button (X) in the top right corner. The background content is partially obscured by the modal. At the bottom of the page, there are 'Previous' and 'Next' navigation buttons.

Activity Lab 3.6

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TensorBoard

Profiler

Partner AI Apps

▼ Model training & customization

Notebooks and Git repos

▼ Try the new JupyterLab in SageMaker Studio

Try the new JupyterLab in SageMaker Studio

- Launch notebooks in seconds and start coding instantly
- Use the similar underlying compute and storage as your notebook instances to enable more features at the same cost
- Seamlessly perform comprehensive ML and analytics workflows, all in one notebook
- Leverage GenAI-powered coding assistance from Amazon Q Developer and JupyterAI to accelerate development
- Collaborate with your peers in real-time on the same notebook for seamless ideation

▶ How to access JupyterLab in Studio?

Notebook instances Info

Find notebook instances

Create notebook instance

Name	Instance	Creation time	Status	Actions
MyNotebook	ml.m4.xlarge	4/29/2026, 1:30:00 AM	InService	Open Jupyter Open JupyterLab

Challenge Task 1

File Edit View Run Kernel Git Tabs Settings Help

3_6-machinelearning.ipynb

False Discovery Rate: 13.6363636363635%
Accuracy: 83.87896774193549%

Challenge task: Record the previous values, then go back to step 1 and change the value used for the threshold. Values you should try are .25 and .75.

Did those threshold values make a difference?

- Yes, the threshold significantly impacts results, lower threshold increases sensitivity while higher threshold increases specificity

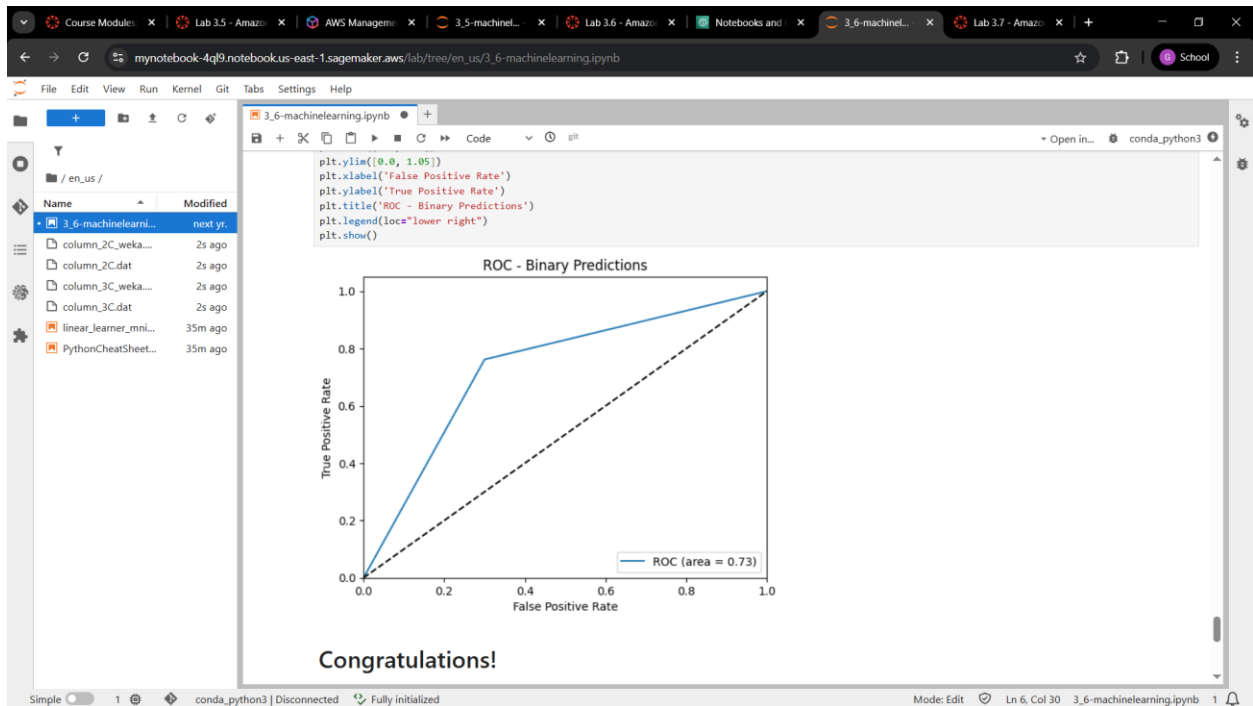
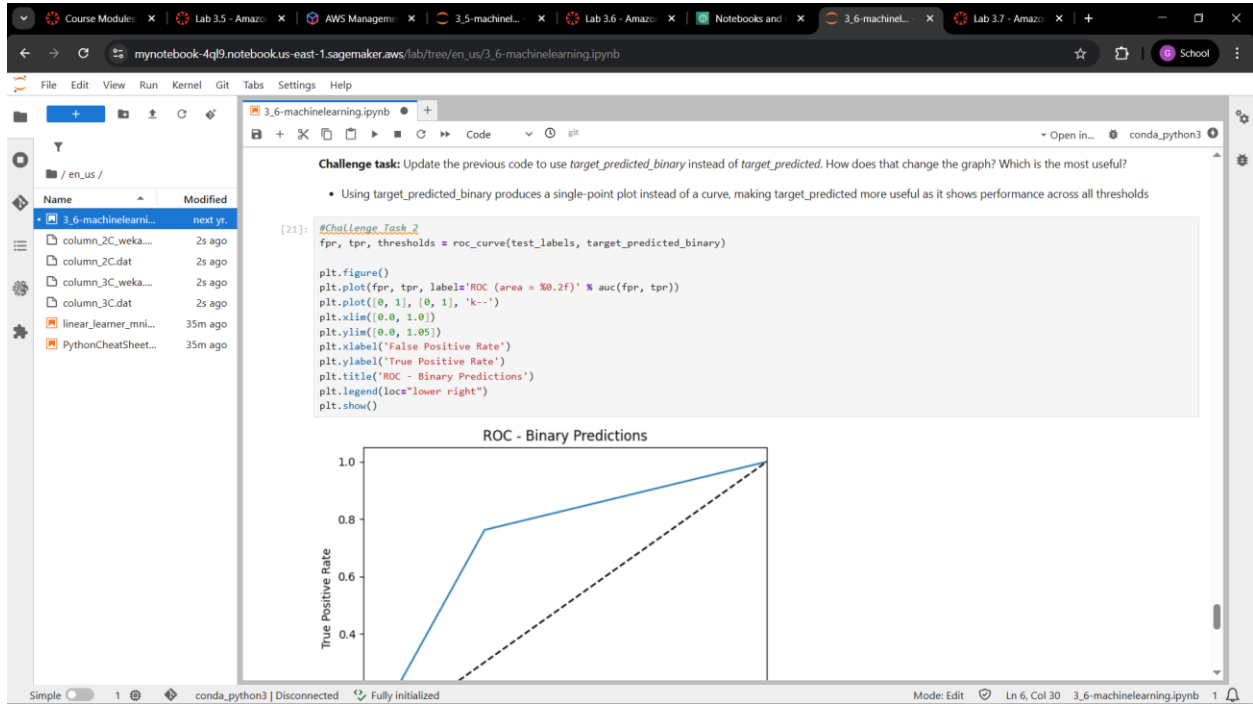
```
[10]: #Challenge_Task_1
for threshold in [0.25, 0.75]:
    def binary_convert(x, tthreshold):
        return 1 if x > t else 0

    col = target_predicted['class']
    target_predicted_binary = col.apply(binary_convert)
    TN, FP, FN, TP = confusion_matrix(
        test_labels, target_predicted_binary
    ).ravel()
    ACC = float((TP+TN)/((TP+FP+FN+TN)*100)
    Sensitivity = float(TP)/((TP+FN)*100)
    Specificity = float(TN)/((TN+FP)*100)
    print(f"Threshold: {threshold}")
    print(f" Accuracy: {ACC:.1f}%")
    print(f" Sensitivity: {Sensitivity:.1f}%")
    print(f" Specificity: {Specificity:.1f}%")

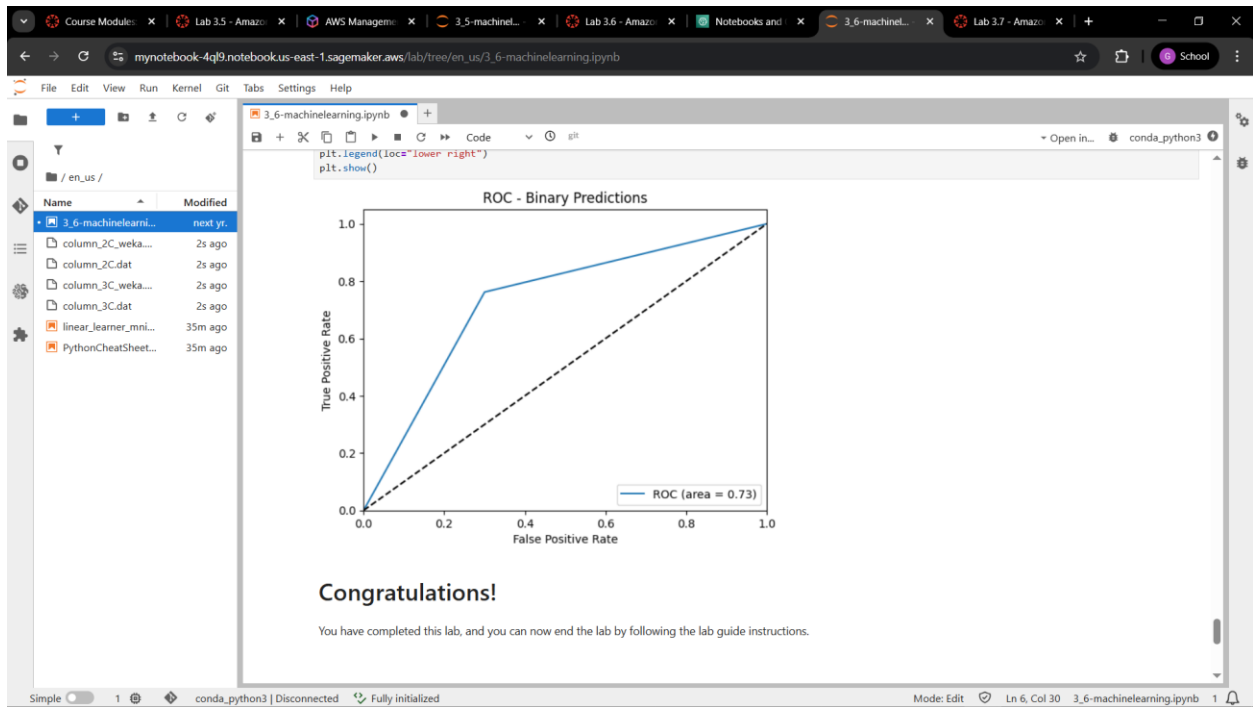
Threshold: 0.25
Accuracy: 80.6%
Sensitivity: 90.5%
Specificity: 60.0%

Threshold: 0.75
Accuracy: 74.2%
Sensitivity: 76.2%
Specificity: 70.0%
```


Challenge Task 2



Last Codes



Canvas Details

Course Modules > Lab 3.5 - Amazon SageMaker > Lab 3.6 - Amazon SageMaker - Generating model performance metrics

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Cloud Access

AWS CLI: Show

Cloud Labs

Remaining session time: 01:47:31 (108 minutes)

Session started at: 2026-04-28T15:08:24-0700

Session to end at: 2026-04-28T17:08:24-0700

Accumulated lab time: 02:12:00 (132 minutes)

No running instance

SSH key: Show Download PEM Download PPK

AWS SSO: Download URL

After completing this lab, you will be able to:

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Activity Lab 3.7

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Dashboard

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Images

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- Leverage GenAI-powered coding assistance from Amazon Q Developer and JupyterAI to accelerate development
- Collaborate with your peers in real-time on the same notebook for seamless ideation

► How to access JupyterLab in Studio?

Notebook instances Info

Find notebook instances

Actions Create notebook instance

Name	Instance	Creation time	Status	Actions
MyNotebook	ml.m4.xlarge	4/29/2026, 6:22:22 AM	InService	Open Jupyter Open JupyterLab

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Last Codes

mynotebook-sagrn.notebook-us-east-1.sagemaker.aws/lab/tree/en_us/3_7-machinelearning.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Launcher 3_7-machinelearning.ipynb

Plot the ROC chart.

```
[15]: plot_roc(test_labels, best_target_predicted_binary)
```

Sensitivity or TPR: 95.23809523809523%
Specificity or TNR: 70.0%
Precision: 86.95652173913044%
Negative Predictive Value: 87.5%
False Positive Rate: 30.0%
False Negative Rate: 4.761904761904762%
False Discovery Rate: 13.043478260869565%
Accuracy: 87.09677419354838%
Validation AUC: 0.8261904761904761

ValueError: Traceback (most recent call last)
Cell In[15], line 1
----> 1 plot_roc(test_labels, best_target_predicted_binary)

File <timed exec>:49, in plot_roc(test_labels, target_predicted_binary)

File ~/anaconda3/envs/python3/lib/python3.12/site-packages/matplotlib/axes/_base.py:4062, in _AxesBase.set_ylim(self, bottom, top, emit, auto, ymin, ymax)

4060 raise TypeError("Cannot pass both 'top' and 'ymax'")
4061 top = ymax
-> 4062 return self.yaxis.set_lim(bottom, top, emit=emit, auto=auto)

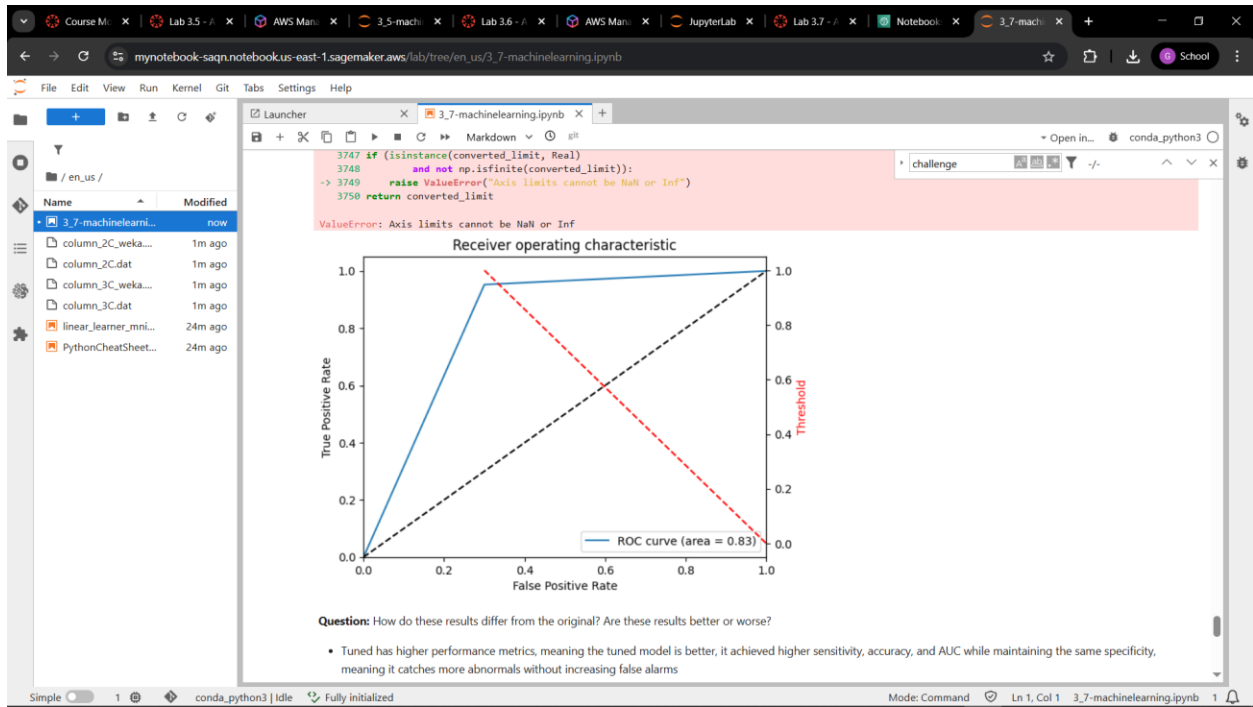
File ~/anaconda3/envs/python3/lib/python3.12/site-packages/matplotlib/axis.py:1217, in Axis.set_lim(self, v0, v1, emit, auto)

1215 self.axes._process_unit_info([(name, (v0, v1))], convert=False)
1216 v0 = self.axes._validate_converted_limits(v0, self.convert_units)
-> 1217 v1 = self.axes._validate_converted_limits(v1, self.convert_units)
1219 if v0 is None or v1 is None:
1220 # Axes init calls set_xlim(0, 1) before get_xlim() can be called,
1221 # so only grab the limits if we really need them.
1222 old0, old1 = self.get_view_interval()

File ~/anaconda3/envs/python3/lib/python3.12/site-packages/matplotlib/axes/_base.py:3749, in _AxesBase._validate_converted_limits(self, limit, convert)

3746 converted_limit = converted_limit.squeeze()

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 1, Col 1 3_7-machinelearning.ipynb 1



Canvas Details

Course M... Lab 3.5 - / AWS Man... 3.5-mach... Lab 3.6 - / AWS Man... JupyterLab Lab 3.7 - / Notebook 3.7-mach... +

awsacademy.instructure.com/courses/168322/modules/items/16494089

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AMLFv1EN-LTI13-168322 > Modules > Module 3 - Implementing a Machine Learning pipeline with Amazon SageMaker > Lab 3.7 - Amazon SageMaker - Hyperparameter Tuning

Details AWS Start Lab End Lab 0:57 Actions

EN-US

Credentials

Cloud Access

AWS CLI: Show

Cloud Labs

Remaining session time: 00:56:13 (57 minutes)
Session started at: 2026-04-28T15:21:46-0700
Session to end at: 2026-04-28T17:21:46-0700

Accumulated lab time: 01:03:00 (63 minutes)

No running instance

SSH key: Show Download PEM Download PPK

AWS SSO: Download URL

In this tune i comp

Obj

After

- Tune an XGBoost model by using Amazon SageMaker

Previous Next

JOSE RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES AND ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CPE C313

ARTIFICIAL INTELLIGENCE

LABORATORY

Challenge

Pozon, Gion Terrence B.

301G

2026-04-29

Challenge Lab

AWS Console account & MyNotebook

The screenshot shows the AWS SageMaker console interface. At the top, a green banner states: "Your notebook instance is being created. Open the notebook instance when status is InService and open a template notebook to get started." Below this, the "Notebook instances" section is visible, featuring a table with the following data:

Name	Instance	Creation time	Status	Actions
my-flight-notebook	m4.xlarge	4/29/2026, 4:28:00 PM	InService	Open Jupyter Open JupyterLab

The left sidebar contains navigation links for "Amazon SageMaker AI", "Environment configuration", "Applications and IDEs", and "Model training & customization".

Code & Output

The screenshot displays the JupyterLab interface with a file explorer on the left and a code editor on the right. The code editor shows the following content:

```
Project presentation: Include a summary of these details in your project presentation.
```

- Determine if and why ML is an appropriate solution to deploy for this scenario.**
#Write your answer here
ML is appropriate because predicting flight delays involves complex relationships among many variables, a supervised ML model can learn these patterns and generalize to unseen flights
- Formulate the business problem, success metrics, and desired ML output.**
#Write your answer here
Business Problem: A travel booking platform wants to notify customers of flight delays at the time of booking
Success Metric: AUC-ROC, F1-score, and Accuracy
Desired ML output: A binary label classification if a flight is delayed or on time
- Identify the type of ML problem that you're working with.**
#Write your answer here
Binary classification using Supervised Learning
- Analyze the appropriateness of the data that you're working with.**
#Write your answer here
The dataset is appropriate: since it has plenty of rows and has appropriate features that are relevant in predicting the desired output

The bottom status bar indicates the environment is "conda_python3 | Idle" and "Fully initialized".

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb

conda_python3

[7]: `df_temp = pd.read_csv(f"{csv_base_path}On_Time_Reporting_Carrier_On-Time_Performance_(1987-present)_2018_9.csv")`

Question: Print the row and column length in the dataset, and print the column names.

Hint: To view the rows and columns of a DataFrame, use the `<DataFrame>.shape` function. To view the column names, use the `<DataFrame>.columns` function.

[8]: `df_shape = df_temp.shape # **ENTER YOUR CODE HERE**
print(f'Rows and columns in one CSV file is {df_shape}')`

Rows and columns in one CSV file is (585749, 110)

Question: Print the first 10 rows of the dataset.

Hint: To print x number of rows, use the built-in `head(x)` function in pandas.

[9]: `# Enter your code here
df_temp.head(10)`

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	DOT_ID_Reporting_Airline	IATA_CODE_Reporting_Airline	Tail_Number	...	Div4TailN
0	2018	3	9	3	1	2018-09-03	9E	20363	9E	N908XJ	...	f
1	2018	3	9	9	7	2018-09-09	9E	20363	9E	N315PQ	...	f
2	2018	3	9	10	1	2018-09-10	9E	20363	9E	N582CA	...	f
3	2018	3	9	13	4	2018-09-13	9E	20363	9E	N292PQ	...	f
4	2018	3	9	14	5	2018-09-14	9E	20363	9E	N600LR	...	f

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 1, Col 26 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

conda_python3

Hint: To print x number of rows, use the built-in `head(x)` function in pandas.

[9]: `# Enter your code here
df_temp.head(10)`

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	DOT_ID_Reporting_Airline	IATA_CODE_Reporting_Airline	Tail_Number	...	Div4TailN
0	2018	3	9	3	1	2018-09-03	9E	20363	9E	N908XJ	...	f
1	2018	3	9	9	7	2018-09-09	9E	20363	9E	N315PQ	...	f
2	2018	3	9	10	1	2018-09-10	9E	20363	9E	N582CA	...	f
3	2018	3	9	13	4	2018-09-13	9E	20363	9E	N292PQ	...	f
4	2018	3	9	14	5	2018-09-14	9E	20363	9E	N600LR	...	f
5	2018	3	9	16	7	2018-09-16	9E	20363	9E	N316PQ	...	f
6	2018	3	9	17	1	2018-09-17	9E	20363	9E	N916XJ	...	f
7	2018	3	9	20	4	2018-09-20	9E	20363	9E	N371CA	...	f
8	2018	3	9	21	5	2018-09-21	9E	20363	9E	N601LR	...	f
9	2018	3	9	23	7	2018-09-23	9E	20363	9E	N906XJ	...	f

10 rows x 110 columns

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 2, Col 17 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb

Open in... conda_python3

Question: Print all the columns in the dataset. To view the column names, use `<DataFrame>.columns`.

```
[10]: print(f'The column names are :')
      print('*****')
      for col in df_temp.columns: # **ENTER YOUR CODE HERE**
          print(col)
```

The column names are :

Year
Quarter
Month
DayOfMonth
DayOfWeek
FlightDate
Reporting_Airline
DOT_ID_Reporting_Airline
IATA_CODE_Reporting_Airline
Tail_Number
Flight_Number_Reporting_Airline
OriginAirportID
OriginAirportSeqID
OriginCityMarketID
Origin
OriginCityName
OriginState
OriginStateFips
OriginStateName
OriginMac
DestAirportID
DestAirportSeqID
DestCityMarketID
Dest
DestCityName
DestState
DestStateFips
DestStateName
DestMac

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 4, Col 15 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb

Open in... conda_python3

For example: `5 in [1,2,3,4,5]`

```
[11]: # Enter your code here
      [col for col in df_temp.columns if 'Del' in col]
```

```
[11]: ['DepDelay',
      'DepDelayMinutes',
      'DepDel15',
      'DepartureDelayGroups',
      'ArrDelay',
      'ArrDelayMinutes',
      'ArrDel15',
      'ArrivalDelayGroups',
      'CarrierDelay',
      'WeatherDelay',
      'NASDelay',
      'SecurityDelay',
      'LateAircraftDelay',
      'DivArrDelay']
```

Here are some more questions to help you learn more about your dataset.

Questions

1. How many rows and columns does the dataset have?
2. How many years are included in the dataset?
3. What is the date range for the dataset?
4. Which airlines are included in the dataset?
5. Which origin and destination airports are covered?

Hints

- To show the dimensions of the DataFrame, use `df_temp.shape`.
- To refer to a specific column, use `df_temp.columnName` (for example, `df_temp.CarrierDelay`).
- To get unique values for a column, use `df_temp.column.unique()` (for example `df_temp.Year.unique()`).

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 4, Col 15 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiaa.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb

conda_python3

Hints

- To show the dimensions of the DataFrame, use `df_temp.shape`.
- To refer to a specific column, use `df_temp.columnName` (for example, `df_temp.CarrierDelay`).
- To get unique values for a column, use `df_temp.column.unique()` (for example `df_temp.Year.unique()`).

```
[12]: print("The rows and #columns are ", df_temp.shape[0], " and ", df_temp.shape[1])
print("The years in this dataset are: ", df_temp.Year.unique())
print("The months covered in this dataset are: ", df_temp.Month.unique())
print("The date range for data is: ", min(df_temp.FlightDate), " to ", max(df_temp.FlightDate))
print("The airlines covered in this dataset are: ", list(df_temp.Reporting_Airline.unique()))
print("The Origin airports covered are: ", list(df_temp.Origin.unique()))
print("The Destination airports covered are: ", list(df_temp.Dest.unique()))
```

The #rows and #columns are 585749 and 110
The years in this dataset are: [2018]
The months covered in this dataset are: [9]
The date range for data is: 2018-09-01 to 2018-09-30
The airlines covered in this dataset are: ['9E', 'B6', 'WN', 'YV', 'YX', 'EV', 'AA', 'AS', 'DL', 'HA', 'UA', 'F9', 'G4', 'MQ', 'NK', 'OH', 'OO']
The Origin airports covered are: ['DFW', 'LGA', 'MSN', 'MSP', 'ATL', 'BOL', 'VLD', 'JFK', 'ROU', 'CHS', 'DTW', 'GRB', 'PVD', 'SHV', 'FNT', 'PIT', 'RI', 'C', 'RST', 'RSW', 'CVG', 'LIT', 'ORD', 'JAX', 'TRI', 'BOS', 'OMA', 'DCA', 'CHO', 'AVP', 'IND', 'GRR', 'BTR', 'MEM', 'TUL', 'CLE', 'STL', 'BTW', 'OMA', 'MHT', 'TVC', 'SAV', 'GSP', 'EWR', 'OAJ', 'BNA', 'MCI', 'TLH', 'ROC', 'LEX', 'PAM', 'BUE', 'AGS', 'CLT', 'GSO', 'BWI', 'SAT', 'PHL', 'TYS', 'ACK', 'DS', 'M', 'GM', 'AVL', 'BGR', 'MHT', 'ILM', 'MDT', 'IAH', 'SBN', 'SYR', 'ORF', 'RKE', 'XNA', 'MSY', 'PBI', 'ABE', 'HPH', 'EVV', 'ALB', 'LNU', 'AUS', 'PHF', 'CHW', 'GTR', 'BWI', 'BGR', 'CID', 'CAK', 'ATW', 'ABY', 'CAE', 'SRQ', 'MLI', 'BHM', 'IAD', 'CSG', 'OMH', 'MCO', 'MBS', 'FLL', 'SOF', 'TPA', 'MFW', 'LAS', 'LGB', 'SFO', 'SAN', 'LAX', 'RNO', 'PDX', 'ANC', 'ABQ', 'SLC', 'DEN', 'PHX', 'OAK', 'SMF', 'SJC', 'SEA', 'HOU', 'STX', 'BUR', 'SWF', 'SJC', 'DAB', 'BQN', 'PSE', 'ORH', 'HYA', 'STT', 'ONT', 'HRL', 'ICT', 'ISP', 'LBB', 'MAF', 'MDW', 'OKC', 'PNS', 'SNA', 'TUS', 'AMA', 'BOI', 'CRP', 'DAL', 'ECP', 'EL', 'P', 'GEG', 'LFT', 'MFE', 'MDT', 'JAN', 'COS', 'MOB', 'VPS', 'MTJ', 'DRO', 'GPT', 'BFL', 'MRV', 'SBA', 'PSP', 'FSD', 'BRO', 'RAP', 'COU', 'STS', 'PIA', 'FAT', 'SBP', 'FSM', 'HSV', 'BIS', 'DAY', 'BZN', 'MIA', 'EYW', 'MYR', 'HHH', 'GJT', 'FAR', 'SGF', 'HOB', 'CLL', 'LRD', 'AEX', 'ERI', 'MLU', 'LCH', 'RO', 'A', 'LAW', 'MHK', 'GRK', 'SAF', 'GRI', 'JLN', 'ROW', 'FWA', 'CRW', 'LAN', 'OGG', 'HNL', 'KOA', 'EGE', 'LIH', 'MLB', 'JAC', 'FAI', 'RDM', 'ADQ', 'BET', 'BRW', 'SCC', 'KTN', 'YAK', 'COV', 'JNU', 'SIT', 'PSG', 'NRG', 'OME', 'OTZ', 'ADK', 'FCA', 'FAY', 'PSC', 'BIL', 'MSO', 'ITO', 'PPG', 'MFR', 'EUG', 'GU', 'H', 'SBN', 'DLH', 'TTH', 'BKG', 'SRB', 'PIE', 'PRO', 'AZA', 'SRX', 'RFD', 'SCK', 'OMB', 'HTS', 'BLU', 'IAG', 'USA', 'GFK', 'BLI', 'ELN', 'PBG', 'LCK', 'GTF', 'OGD', 'IDA', 'PVU', 'TOL', 'PSM', 'CKB', 'HGR', 'SPI', 'STC', 'ACT', 'TYR', 'ABI', 'AZO', 'CHI', 'BPT', 'GCK', 'HQT', 'ALO', 'TXK', 'SPS', 'SW', 'O', 'DRQ', 'SUX', 'SJT', 'GGG', 'LSE', 'LBE', 'ACY', 'LYH', 'PGV', 'HVN', 'ENN', 'DHN', 'PIH', 'INT', 'WYS', 'CPR', 'SCE', 'HNL', 'SUN', 'ISN', 'CHK', 'EAU', 'LNB', 'SHD', 'LBF', 'HYS', 'SLN', 'EAR', 'VEL', 'CHY', 'GCC', 'RKS', 'PUB', 'LBL', 'MKG', 'PAH', 'CGI', 'UIN', 'BFF', 'DVL', 'JMS', 'LAR', 'SG', 'U', 'PRC', 'ASE', 'RDD', 'ACV', 'OTH', 'COD', 'LWS', 'ABR', 'APN', 'ESC', 'PLN', 'BII', 'BRD', 'BTM', 'CDC', 'CIU', 'EKO', 'TWF', 'HIB', 'BGM', 'RHI', 'ITH', 'INL', 'FLG', 'YUM', 'HEI', 'PIB', 'HDN']
The Destination airports covered are: ['CVG', 'PAM', 'RNO', 'HSP', 'MSN', 'SHV', 'CLT', 'PIT', 'BTR', 'YAH', 'ATI', 'TEK', 'DCA', 'DTW', 'LGA', 'TYS', '...

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 4, Col 15 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

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documentation).

```
[13]: counts = pd.DataFrame({'Origin':df_temp.Origin.value_counts(), 'Destination':df_temp.Dest.value_counts()})
counts
```

```
[13]:
```

Origin	Destination
ABE	303
ABI	169
ABQ	2077
ABR	60
ABY	79
...	...
WRG	60
WYS	52
XNA	1004
YAK	60
YUM	96

346 rows x 2 columns

Question: Print the top 15 origin and destination airports based on number of flights in the dataset.

Hint: You can use the `sort_values` function in pandas (pandas.DataFrame.sort_values documentation).

```
[14]: counts.sort_values(by='Origin', ascending=False).head(15) # Enter your code here
```

```
[14]:
```

Origin	Destination
--------	-------------

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Question: Print the top 15 origin and destination airports based on number of flights in the dataset.

Hint: You can use the `sort_values` function in pandas (pandas.DataFrame.sort_values documentation).

```
[14]: counts.sort_values(by='Origin', ascending=False).head(15) # Enter your code here
```

```
[14]:
```

	Origin	Destination
ATL	31525	31521
ORD	28257	28250
DFW	22802	22795
DEN	19807	19807
CLT	19655	19654
LAX	17875	17873
SFO	14332	14348
IAH	14210	14203
LGA	13850	13850
MSP	13349	13347
LAS	13318	13322
PHX	13126	13128
DTW	12725	12724
BOS	12223	12227
SEA	11872	11877

Given all the information about a flight trip, can you predict if it would be delayed?

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Flight_Delay-Student.ipynb

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```
[18]: data = pd.read_csv(combined_csv_filename)
```

Print the first five records.

```
[19]: # Enter your code here
data.head()
```

```
[19]:
```

	Year	Quarter	Month	DayofMonth	DayOfWeek	FlightDate	Reporting_Airline	Origin	OriginState	Dest	DestState	CRSDepTime	Cancelled	Diverted	Distance
0	2015	4	12	1	2	2015-12-01	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
1	2015	4	12	2	3	2015-12-02	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
2	2015	4	12	3	4	2015-12-03	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
3	2015	4	12	4	5	2015-12-04	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0
4	2015	4	12	5	6	2015-12-05	AA	SFO	CA	DFW	TX	819	0.0	0.0	1464.0

Here are some more questions to help you learn more about your dataset.

Questions

1. How many rows and columns does the dataset have?
2. How many years are included in the dataset?
3. What is the date range for the dataset?
4. Which airlines are included in the dataset?
5. Which origin and destination airports are covered?

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4. Which airlines are included in the dataset?
5. Which origin and destination airports are covered?

```
[20]: print("The #rows and #columns are:", data.shape[0], " and ", data.shape[1])
print("The years in this dataset are: ", list(data.Year.unique()))
print("The months covered in this dataset are: ", sorted(list(data.Month.unique())))
print("The date range for data is: ", min(data.FlightDate), " to ", max(data.FlightDate))
print("The airlines covered in this dataset are: ", list(data.Reporting_Airline.unique()))
print("The Origin airports covered are: ", list(data.Origin.unique()))
print("The Destination airports covered are: ", list(data.Dest.unique()))
```

The #rows and #columns are 1658130 and 20
The years in this dataset are: [2015, 2014, 2017, 2016, 2018]
The months covered in this dataset are: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
The date range for data is: 2014-01-01 to 2018-12-31
The airlines covered in this dataset are: ['AA', 'DL', 'OO', 'UA', 'WN']
The Origin airports covered are: ['SFO', 'DFW', 'LAX', 'PHX', 'DEN', 'ORD', 'CLT', 'ATL', 'IAH']
The Destination airports covered are: ['DFW', 'SFO', 'ORD', 'LAX', 'PHX', 'ATL', 'IAH', 'CLT', 'DEN']

Define your target column: **is_delay** (1 means that the arrival time delayed more than 15 minutes, and 0 means all other cases). To rename the column from **ArrDel15** to **is_delay**, use the `rename` method.

Hint: You can use the `rename` function in pandas ([pandas.DataFrame.rename documentation](#)).

For example:

```
data.rename(columns={'col1':'column1'}, inplace=True)
```

```
[21]: data.rename(columns={'ArrDel15': 'is_delay'}, inplace=True) # Enter your code here
```

Look for nulls across columns. You can use the `isnull()` function ([pandas.isnull documentation](#)).

Hint: `isnull()` detects whether the particular value is null or not. It returns a boolean (*True* or *False*) in its place. To sum the number of columns, use the `sum(axis=0)` function (for example, `df.isnull().sum(axis=0)`).

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```
[21]: data.rename(columns={'ArrDel15': 'is_delay'}, inplace=True) # Enter your code here
```

Look for nulls across columns. You can use the `isnull()` function ([pandas.isnull documentation](#)).

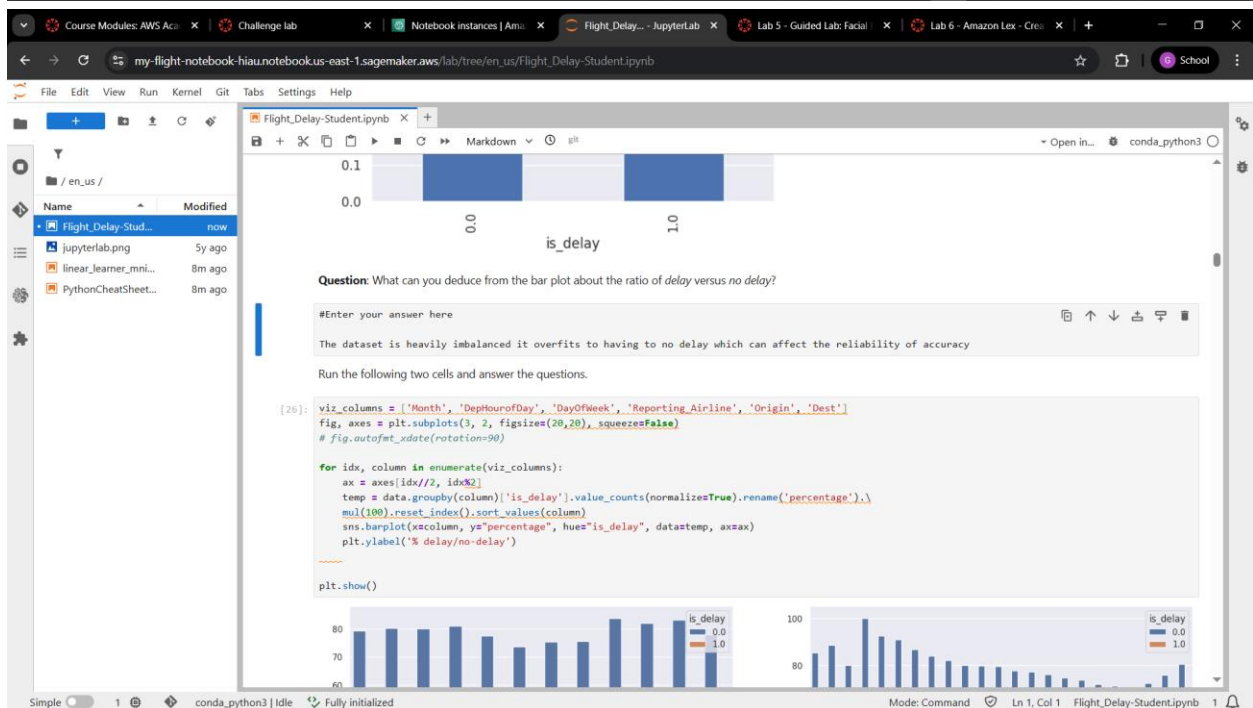
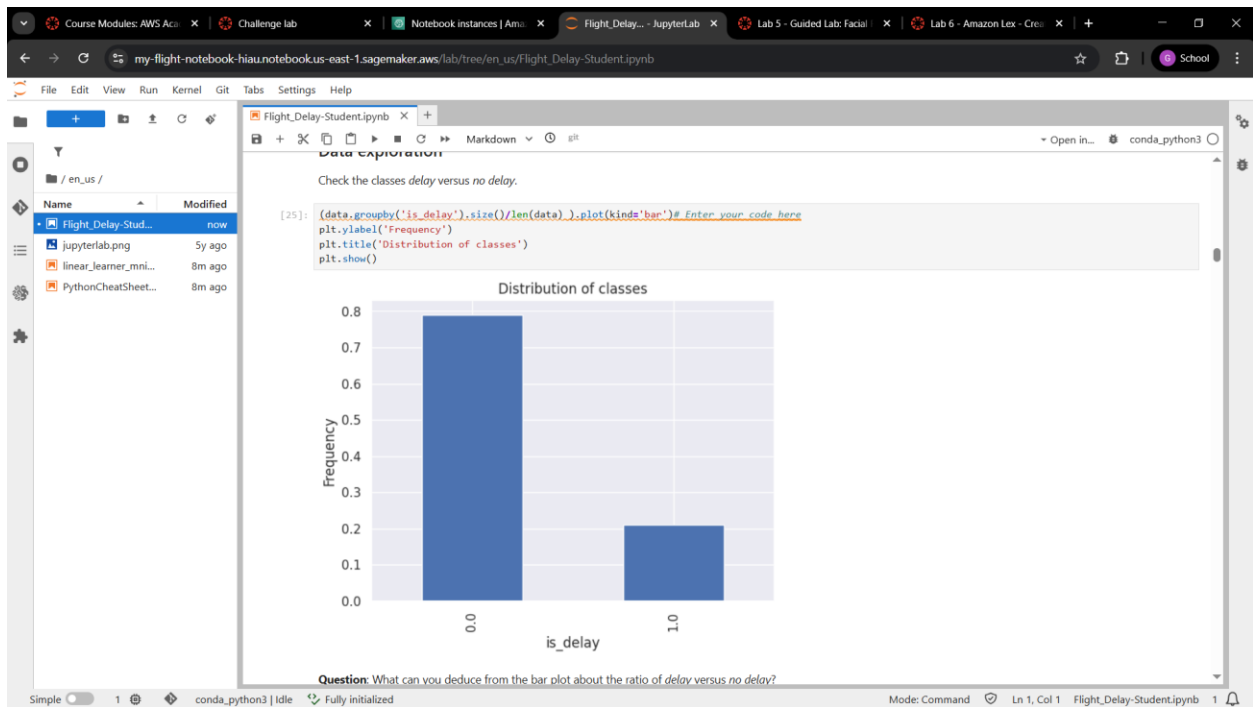
Hint: `isnull()` detects whether the particular value is null or not. It returns a boolean (*True* or *False*) in its place. To sum the number of columns, use the `sum(axis=0)` function (for example, `df.isnull().sum(axis=0)`).

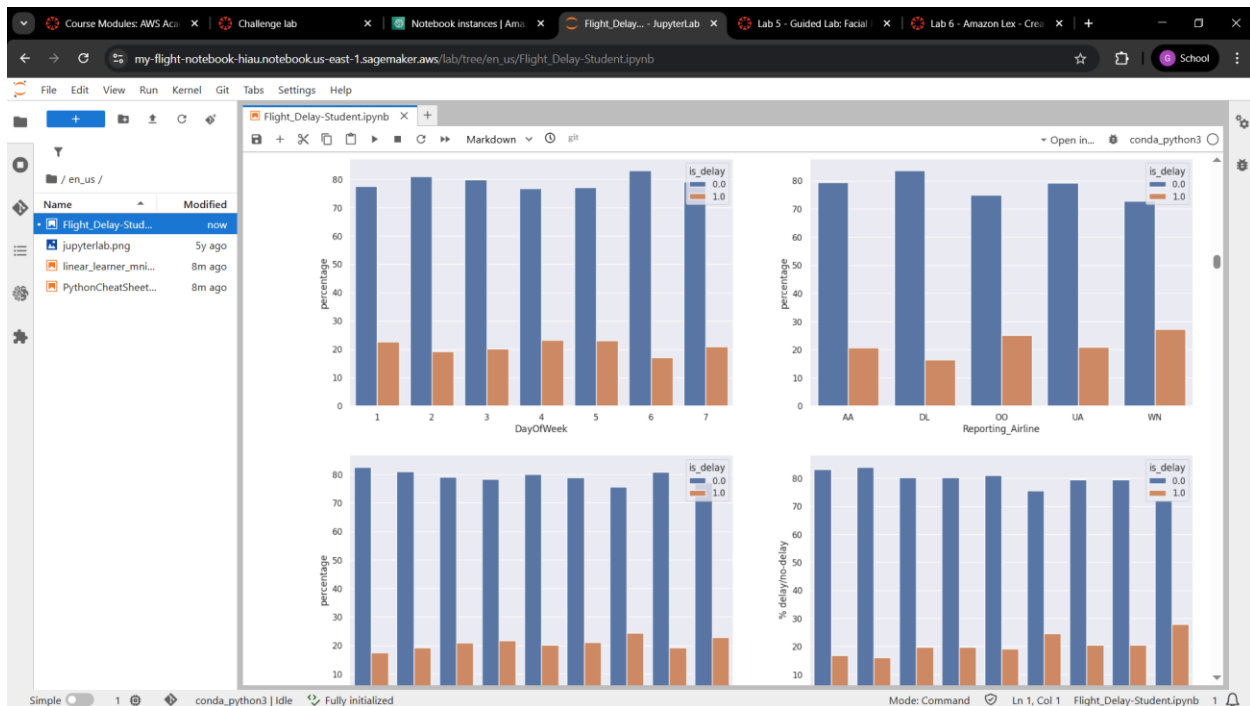
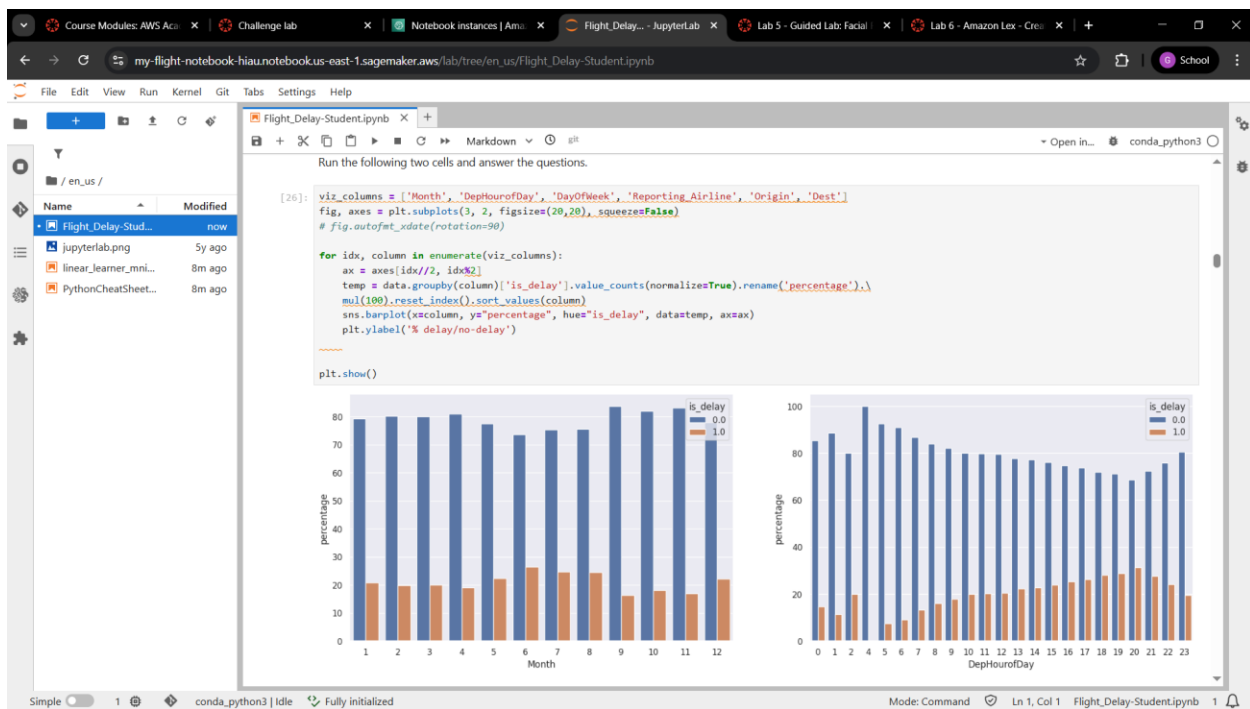
```
[22]: # Enter your code here
data.isnull().sum(axis=0)
```

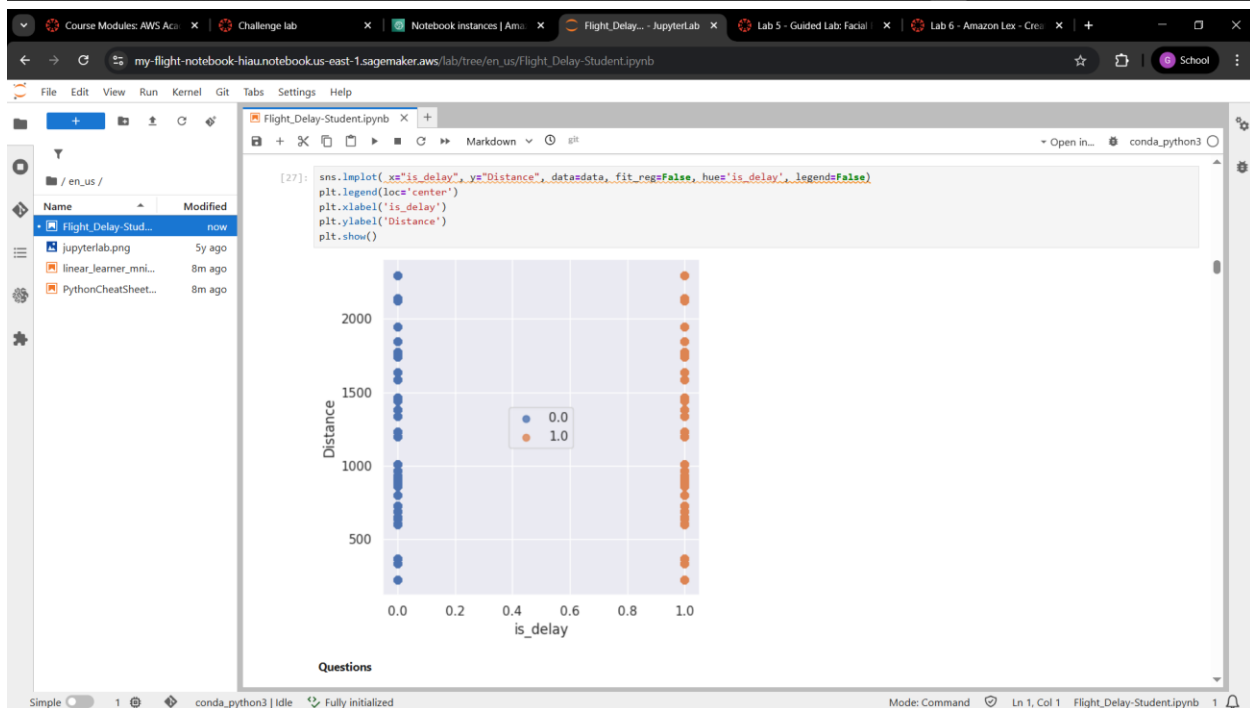
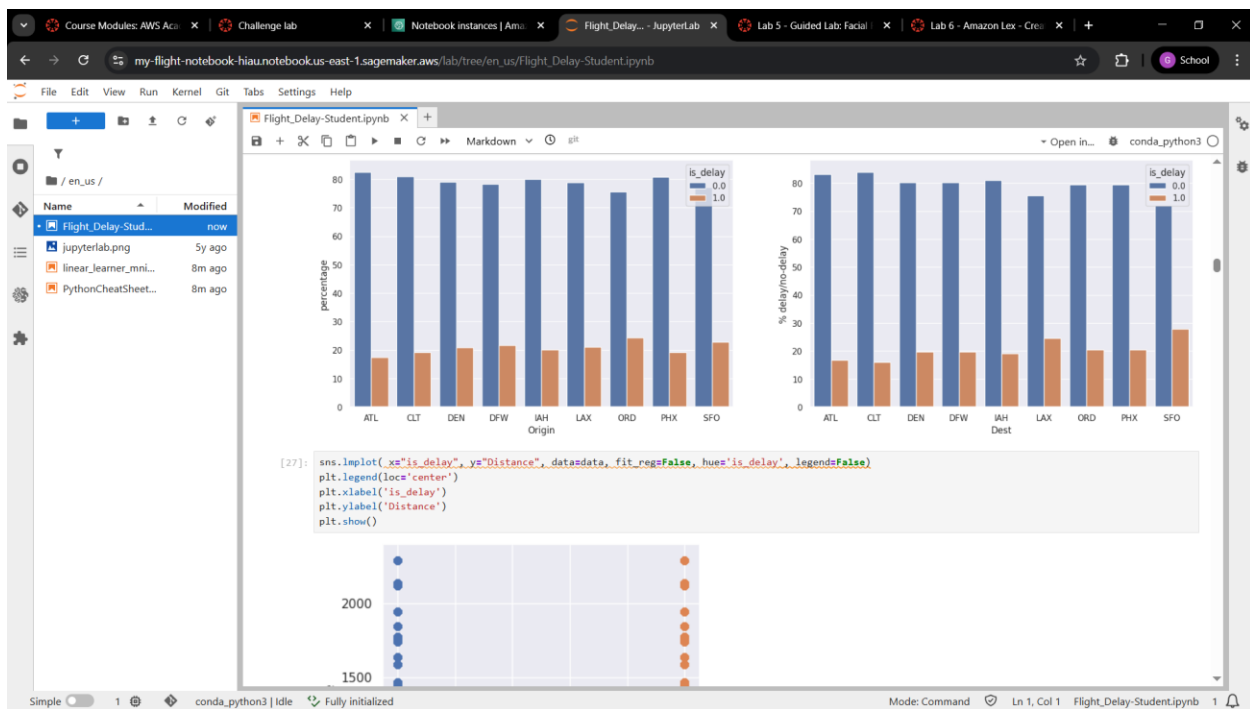
```
[22]: Year      0
Quarter      0
Month        0
DayofMonth   0
DayOfWeek    0
FlightDate    0
Reporting_Airline  0
Origin        0
OriginState   0
Dest          0
DestState     0
CRSDepTime    0
Cancelled     0
Diverted      0
Distance      0
DistanceGroup 0
ArrDelay      22540
ArrDelayMinutes 22540
is_delay      22540
AirTime       22540
dtype: int64
```

The arrival delay details and airtime are missing for 22,540 out of 1,658,130 rows, which is 1.3 percent. You can either remove or impute these rows. The documentation doesn't mention any information about missing rows.

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Flight_Delay-Student.ipynb x +

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Questions

Using the data from the previous charts, answer these questions:

- Which months have the most delays?
- What time of the day has the most delays?
- What day of the week has the most delays?
- Which airline has the most delays?
- Which origin and destination airports have the most delays?
- Is flight distance a factor in the delays?

#Enter your answers here

- Most delayed months: May, June, July, December
- Most delayed time of day: Evening at 8 PM
- Most delayed day of week: Monday, Thursday, Friday
- Most delayed airline: OO and WN
- Most delayed airports: ORD
- Distance: no since the values are similar with each other indicating little to no impact as a factor in predicting delays

Features

Look at all the columns and what their specific types are.

```
[28]: data.columns
```

```
[28]: Index(['Year', 'Quarter', 'Month', 'DayOfMonth', 'DayOfWeek', 'FlightDate',  
       'Reporting_Airline', 'Origin', 'OriginState', 'Dest', 'DestState',  
       'CRSDepTime', 'Cancelled', 'Diverted', 'Distance', 'DistanceGroup',  
       'ArrDelay', 'ArrDelayMinutes', 'is_delay', 'AirTime', 'DepHourOfDay'],  
      dtype='object')
```

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Course Modules: AWS Acc... x Challenge lab x Notebook instances | Ami... x Flight_Delay... - JupyterLab x Lab 5 - Guided Lab: Facial x Lab 6 - Amazon Lex - Cre... x +

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```
[31]: data_dummies = pd.get_dummies(data[categorical_columns], drop_first=True) # Enter your code here  
data_dummies = data_dummies.replace({True: 1, False: 0})  
data = pd.concat([data, data_dummies], axis = 1)  
data.drop(categorical_columns,axis=1,inplace=True)
```

Check the length of the dataset and the new columns.

Hint Use the `shape` and `columns` properties.

```
[32]: # Enter your code here  
data.shape
```

```
[32]: (1635598, 94)
```

```
[33]: # Enter your code here  
data.columns
```

```
[33]: Index(['is_delay', 'Distance', 'Quarter_2', 'Quarter_3', 'Quarter_4',  
       'Month_2', 'Month_3', 'Month_4', 'Month_5', 'Month_6', 'Month_7',  
       'Month_8', 'Month_9', 'Month_10', 'Month_11', 'Month_12',  
       'DayOfMonth_2', 'DayOfMonth_3', 'DayOfMonth_4', 'DayOfMonth_5',  
       'DayOfMonth_6', 'DayOfMonth_7', 'DayOfMonth_8', 'DayOfMonth_9',  
       'DayOfMonth_10', 'DayOfMonth_11', 'DayOfMonth_12', 'DayOfMonth_13',  
       'DayOfMonth_14', 'DayOfMonth_15', 'DayOfMonth_16', 'DayOfMonth_17',  
       'DayOfMonth_18', 'DayOfMonth_19', 'DayOfMonth_20', 'DayOfMonth_21',  
       'DayOfMonth_22', 'DayOfMonth_23', 'DayOfMonth_24', 'DayOfMonth_25',  
       'DayOfMonth_26', 'DayOfMonth_27', 'DayOfMonth_28', 'DayOfMonth_29',  
       'DayOfMonth_30', 'DayOfMonth_31', 'DayOfWeek_2', 'DayOfWeek_3',  
       'DayOfWeek_4', 'DayOfWeek_5', 'DayOfWeek_6', 'DayOfWeek_7',  
       'Reporting_Airline_DL', 'Reporting_Airline_OO', 'Reporting_Airline_UA',  
       'Reporting_Airline_WN', 'Origin_CLT', 'Origin_DEN', 'Origin_DFW',  
       'Origin_IAH', 'Origin_LAX', 'Origin_ORD', 'Origin_PHX', 'Origin_SFO',  
       'Dest_CLT', 'Dest_DEN', 'Dest_DFW', 'Dest_IAH', 'Dest_LAX', 'Dest_ORD',  
       'Dest_PHX', 'Dest_SFO', 'DepHourOfDay_1', 'DepHourOfDay_2',  
       'DepHourOfDay_4', 'DepHourOfDay_5', 'DepHourOfDay_6', 'DepHourOfDay_7',  
       'DepHourOfDay_8', 'DepHourOfDay_9', 'DepHourOfDay_10',  
       'DepHourOfDay_11', 'DepHourOfDay_12', 'DepHourOfDay_13', 'DepHourOfDay_14',  
       'DepHourOfDay_15', 'DepHourOfDay_16', 'DepHourOfDay_17', 'DepHourOfDay_18',  
       'DepHourOfDay_19', 'DepHourOfDay_20', 'DepHourOfDay_21', 'DepHourOfDay_22',  
       'DepHourOfDay_23', 'DepHourOfDay_24', 'DepHourOfDay_25', 'DepHourOfDay_26',  
       'DepHourOfDay_27', 'DepHourOfDay_28', 'DepHourOfDay_29', 'DepHourOfDay_30',  
       'DepHourOfDay_31'],  
      dtype='object')
```

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```
'DepHourOfDay_17', 'DepHourOfDay_18', 'DepHourOfDay_19',
'DepHourOfDay_20', 'DepHourOfDay_21', 'DepHourOfDay_22',
'DepHourOfDay_23'],
dtype='object')
```

You are now ready to train the model. Before you split the data, rename the **is_delay** column to **target**.

Hint You can use the `rename` function in pandas ([pandas.DataFrame.rename](#) documentation).

```
[34]: data.rename(columns = {'is_delay': 'target'}, inplace=True) # Enter your code here
```

End of Step 2

Save the project file to your local computer. Follow these steps:

1. In the file explorer on the left, right-click the notebook that you're working on.
2. Choose **Download**, and save the file locally.

This action downloads the current notebook to the default download folder on your computer.

Step 3: Model training and evaluation

You must include some preliminary steps when you convert the dataset from a DataFrame to a format that a machine learning algorithm can use. For Amazon SageMaker, you must perform these steps:

1. Split the data into `train_data`, `validation_data`, and `test_data` by using `sklearn.model_selection.train_test_split`.
2. Convert the dataset to an appropriate file format that the Amazon SageMaker training job can use. This can be either a CSV file or record protobuf. For more information, see [Common Data Formats for Training](#).
3. Upload the data to your S3 bucket. If you haven't created one before, see [Create a Bucket](#).

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Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

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Baseline classification model

```
[37]: import sagemaker
from sagemaker.serializers import CSVSerializer
from sagemaker.amazon.amazon_estimator import RecordSet
import boto3

# Instantiate the linearlearner estimator object with 1 ml.m4.xlarge
classifier_estimator = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
                                             instance_count=1,
                                             instance_type='ml.m4.xlarge',
                                             predictor_type='binary_classifier',
                                             binary_classifier_model_selection_criteria='cross_entropy_loss')

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
```

Sample code

```
num_classes = len(pd.unique(train_labels))
classifier_estimator = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
                                             instance_count=1,
                                             instance_type='ml.m4.xlarge',
                                             predictor_type='binary_classifier',
                                             binary_classifier_model_selection_criteria = 'cross_entropy_loss')
```

Linear learner accepts training data in protobuf or CSV content types. It also accepts inference requests in protobuf, CSV, or JavaScript Object Notation (JSON) content types. Training data has features and ground-truth labels, but the data in an inference request has only features.

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Flight_Delay-Student.ipynb x +

Sample code

```
linear.fit([train_records,val_records,test_records])
```

[39]: `### Fit the classifier`
`# Enter your code here`
`classifier_estimator.fit([train_records, val_records, test_records])`

INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker:Creating training job with name: linear-learner-2026-04-29-11-15-09-424
 2026-04-29 11:15:09 Starting - Starting the training job.....
 2026-04-29 11:16:08 Starting - Preparing the instances for training.....
 2026-04-29 11:16:55 Downloading - Downloading input data...
 2026-04-29 11:17:35 Downloading - Downloading the training image.....
 2026-04-29 11:19:01 Training - Training image download completed. Training in progress.. Docker entrypoint called with argument(s): train
 Running default environment configuration script
 [04/29/2026 11:19:11 INFO 14066818178176] Reading default configuration from /opt/amazon/lib/python3.8/site-packages/algorithm/resources/default-input.json: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'no_realize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'auto', 'num_point_for_scaler': '10000', 'kystore': 'auto', 'num_gpus': 'auto', 'num_kv_servers': 'auto', 'log_level': 'info', 'tuning_objective_metric': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.001', 'enable_profiler': 'false'}
 [04/29/2026 11:19:11 INFO 14066818178176] Merging with provided configuration from /opt/ml/input/config/hyperparameters.json: {'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'feature_dim': '93', 'mini_batch_size': '1000', 'predictor_type': 'binary_classifier'}
 [04/29/2026 11:19:11 INFO 14066818178176] Final configuration: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': '93', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'auto'}

Course Modules: AWS Acc... x Challenge lab x Notebook instances | Ami... x Flight_Delay... - JupyterLab x Lab 5 - Guided Lab: Facial x Lab 6 - Amazon Lex - Cre... x +

my-flight-notebook-hiaa.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb x +

To plot the confusion matrix call the `plot_confusion_matrix` function on the `test_labels` and the `target_predicted` data from your batch job:

[46]: `# Enter your code here`
`plot_confusion_matrix(test_labels, target_predicted)`

Confusion Matrix

	0	1
True Class 0	129144.00	82.00
True Class 1	34190.00	143.00

0 1

Predicted Class

Key questions to consider:

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebookus-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

Key questions to consider:

1. How does your model's performance on the test set compare to its performance on the training set? What can you deduce from this comparison?
2. Are there obvious differences between the outcomes of metrics like accuracy, precision, and recall? If so, why might you be seeing those differences?
3. Given your business situation and goals, which metric (or metrics) is the most important for you to consider? Why?
4. From a business standpoint, is the outcome for the metric (or metrics) that you consider to be the most important sufficient for what you need? If not, what are some things you might change in your next iteration? (This will happen in the feature engineering section, which is next.)

Use the following cells to answer these (and other) questions. Insert and delete cells where needed.

Project presentation: In your project presentation, write down your answers to these questions -- and other similar questions that you might answer -- in this section. Record the key details and decisions that you made.

Question: What can you summarize from the confusion matrix?

#Enter your answer here

The model is heavily biased toward predicting no delay, giving it very poor recall on the delay class despite high overall accuracy

End of Step 3

Save the project file to your local computer. Follow these steps:

1. In the file explorer on the left, right-click the notebook that you're working on.
2. Select **Download**, and save the file locally.

This action downloads the current notebook to the default download folder on your computer.

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebookus-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

Because the list of holidays from 2014 to 2018 is known, you can create an indicator variable **is_holiday** to mark them.

The hypothesis is that airplane delays could be higher during holidays compared to the rest of the days. Add a boolean variable **is_holiday** that includes the holidays for the years 2014-2018.

```
[47]: # Source: http://www.calendarpedia.com/holidays/federal-holidays-2014.html
holidays_14 = ['2014-01-01', '2014-01-20', '2014-02-17', '2014-05-26', '2014-07-04', '2014-09-01', '2014-10-13', '2014-11-11', '2014-11-27', '2014-12-25']
holidays_15 = ['2015-01-01', '2015-01-19', '2015-02-16', '2015-05-25', '2015-06-03', '2015-07-04', '2015-09-07', '2015-10-12', '2015-11-11', '2015-11-27', '2015-12-25']
holidays_16 = ['2016-01-01', '2016-01-18', '2016-02-15', '2016-05-30', '2016-07-04', '2016-09-05', '2016-10-10', '2016-11-11', '2016-11-24', '2016-12-25']
holidays_17 = ['2017-01-02', '2017-01-16', '2017-02-20', '2017-05-29', '2017-07-04', '2017-09-04', '2017-10-09', '2017-11-10', '2017-11-23', '2017-12-25']
holidays_18 = ['2018-01-01', '2018-01-15', '2018-02-19', '2018-05-28', '2018-07-04', '2018-09-03', '2018-10-08', '2018-11-12', '2018-11-22', '2018-12-25']
holidays = holidays_14 + holidays_15 + holidays_16 + holidays_17 + holidays_18

## Add indicator variable for holidays
data_orig['is_holiday'] = data_orig['FlightDate'].isin(holidays).astype(int) # Enter your code here
```

Weather data was fetched from <https://www.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00023174,USW00012960,USW00003017,USW00094846,USW00013874,USW00023234,USW00003927,USW00023183,USW00013881&dataTypes=AWT01-01&endDate=2018-12-31>.

This dataset has information on wind speed, precipitation, snow, and temperature for cities by their airport codes.

Question: Could bad weather because of rain, heavy winds, or snow lead to airplane delays? You will now check.

```
[48]: !aws s3 cp s3://aws-tc-largeobjects/CUR-TF-200-ACMLFO-1/flight_delay_project/data2/daily-summaries.csv /home/ec2-user/SageMaker/project/data/
!wget "https://www.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00023174,USW00012960,USW00003017,USW00094846,USW00013874,USW00023234,USW00003927,USW00023183,USW00013881&dataTypes=AWT01-01&endDate=2018-12-31"
download: s3://aws-tc-largeobjects/CUR-TF-200-ACMLFO-1/flight_delay_project/data2/daily-summaries.csv to ../project/data/daily-summaries.csv
```

Import the weather data that was prepared for the airport codes in the dataset. Use the following stations and airports for the analysis. Create a new column called **airport**

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my-flight-notebook-hiaa.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

MONTH 0
dtype: int64

Question: Print the index of the rows that have missing values for TAVG, TMAX, TMIN.

Hint: To find the rows that are missing, use the `isna()` function. Then, to get the index, use the list on the `idx` variable.

```
[52]: idx = np.array([i for i in range(len(weather))])
      TAVG_idx = idx[weather.TAVG.isna()]
      TMAX_idx = idx[weather.TMAX.isna()] # Enter your code here
      TMIN_idx = idx[weather.TMIN.isna()] # Enter your code here
      TAVG_idx
```

```
[52]: array([ 3956, 3957, 3958, 3959, 3960, 3961, 3962, 3963, 3964,
        3965, 3966, 3967, 3968, 3969, 3970, 3971, 3972, 3973,
        3974, 3975, 3976, 3977, 3978, 3979, 3980, 3981, 3982,
        3983, 3984, 3985, 4017, 4018, 4019, 4020, 4021, 4022,
        4023, 4024, 4025, 4026, 4027, 4028, 4029, 4030, 4031,
        4032, 4033, 4034, 4035, 4036, 4037, 4038, 4039, 4040,
        4041, 4042, 4043, 4044, 4045, 4046, 4047, 13420])
```

Sample output

```
array([ 3956, 3957, 3958, 3959, 3960, 3961, 3962, 3963, 3964,
        3965, 3966, 3967, 3968, 3969, 3970, 3971, 3972, 3973,
        3974, 3975, 3976, 3977, 3978, 3979, 3980, 3981, 3982,
        3983, 3984, 3985, 4017, 4018, 4019, 4020, 4021, 4022,
        4023, 4024, 4025, 4026, 4027, 4028, 4029, 4030, 4031,
        4032, 4033, 4034, 4035, 4036, 4037, 4038, 4039, 4040,
        4041, 4042, 4043, 4044, 4045, 4046, 4047, 13420])
```

You can replace the missing TAVG, TMAX, and TMIN values with the average value for a particular station or airport. Because consecutive rows of TAVG_idx are missing, replacing them with a previous value would not be possible. Instead, replace them with the mean. Use the `groupby` function to aggregate the variables with a mean value.

Hint: Group by MONTH and STATION.

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 3, Col 57 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiaa.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

You can replace the missing TAVG, TMAX, and TMIN values with the average value for a particular station or airport. Because consecutive rows of TAVG_idx are missing, replacing them with a previous value would not be possible. Instead, replace them with the mean. Use the `groupby` function to aggregate the variables with a mean value.

Hint: Group by MONTH and STATION.

```
[53]: weather_impute = weather.groupby(['MONTH', 'STATION']).agg({'TAVG': 'mean', 'TMAX': 'mean', 'TMIN': 'mean'}).reset_index() # Enter your code here
      weather_impute.head(2)
```

```
[53]:   MONTH  STATION  TAVG  TMAX  TMIN
0      01  USW00003017 -2.741935  74.000000 -69.858065
1      01  USW00003927  79.529032  143.767742  20.696774
```

Merge the mean data with the weather data.

```
[54]: weather = pd.merge(weather, weather_impute, how='left', left_on=['MONTH', 'STATION'], right_on=['MONTH', 'STATION'])
      .rename(columns = {'TAVG_y': 'TAVG_AVG',
                        'TMAX_y': 'TMAX_AVG',
                        'TMIN_y': 'TMIN_AVG',
                        'TAVG_x': 'TAVG',
                        'TMAX_x': 'TMAX',
                        'TMIN_x': 'TMIN'})
```

Check for missing values again.

```
[55]: weather[TAVG_idx] = weather.TAVG_AVG[TAVG_idx]
      weather[TMAX_idx] = weather.TMAX_AVG[TMAX_idx]
      weather[TMIN_idx] = weather.TMIN_AVG[TMIN_idx]
      weather.isna().sum()
```

```
[55]: STATION 0
      DATE 0
```

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 2, Col 23 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebookus-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

Open in... conda_python3

en_us /

Name	Modified
Flight_Delay-Stud...	now
jupyterlab.png	5y ago
linear_learner_mni...	8m ago
PythonCheatSheet...	8m ago

Rename the `is_delay` column to `target` again. Use the same code that you used previously.

```
[64]: data.rename(columns = {'is_delay': 'target'}, inplace=True) # Enter your code here
```

Create the training sets again.

Hint: Use the `split_data` function that you defined (and used) earlier.

```
[65]: # Enter your code here
train, validate, test = split_data(data)
```

New baseline classifier

Now, see if these new features add any predictive power to the model.

```
[66]: # Instantiate the LinearLearner estimator object
classifier_estimator2 = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
                                              instance_count=1,
                                              instance_types='ml.m4.xlarge',
                                              predictor_types='binary_classifier',
                                              binary_classifier_model_selection_criteria='cross_entropy_loss') # Enter your code here
```

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml

Sample code

```
num_classes = len(pd.unique(train_labels))
classifier_estimator2 = sagemaker.LinearLearner(role=sagemaker.get_execution_role(),
                                              instance_count=1,
                                              instance_types='ml.m4.xlarge')
```

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my-flight-notebook-hiau.notebookus-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

Open in... conda_python3

en_us /

Name	Modified
Flight_Delay-Stud...	now
jupyterlab.png	5y ago
linear_learner_mni...	8m ago
PythonCheatSheet...	8m ago

```
[67]: train_records = classifier_estimator2.record_set(train.values[:, 1:].astype(np.float32), train.values[:, 0].astype(np.float32), channel='train')
val_records = classifier_estimator2.record_set(validate.values[:, 1:].astype(np.float32), validate.values[:, 0].astype(np.float32), channel='validation')
test_records = classifier_estimator2.record_set(test.values[:, 1:].astype(np.float32), test.values[:, 0].astype(np.float32), channel='test')
```

Train your model by using the three datasets that you just created.

```
[68]: # Enter your code here
classifier_estimator2.fit([train_records, val_records, test_records])
```

INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker:Creating training job with name: linear-learner-2026-04-29-11-33-24-846
2026-04-29 11:33:25 Starting - Starting the training job...
2026-04-29 11:33:41 Starting - Preparing the instances for training...
2026-04-29 11:34:05 Downloading - Downloading input data...
2026-04-29 11:34:51 Downloading - Downloading the training image.....
2026-04-29 11:36:07 Training - Training image download completed. Training in progress..Docker entrypoint called with argument(s): train
Running default environment configuration script
[04/29/2026 11:36:21 INFO 140654333134656] Reading default configuration from /opt/amazon/lib/python3.8/site-packages/algorithms/resources/default-input.json: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'no_realize_data': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'auto', 'num_point_for_scaler': '10000', 'kvstore': 'auto', 'num_gpus': 'auto', 'num_kv_servers': 'auto', 'log_level': 'info', 'tuning_objective_metric': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.001', 'enable_profiler': 'false'}
[04/29/2026 11:36:21 INFO 140654333134656] Merging with provided configuration from /opt/ml/input/config/hyperparameters.json: {'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'feature_dim': '85', 'mini_batch_size': '1000', 'predictor_type': 'binary_classifier'}
[04/29/2026 11:36:21 INFO 140654333134656] Final configuration: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': '85', 'use_bias': 'true', 'binary_classifier_model_selection_criteria': 'cross_entropy_loss', 'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto'}

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 1, Col 23 Flight_Delay-Student.ipynb 1

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

conda_python3

```

"training"), "Metrics": [{"initialize.time": {"sum": 279.0954113006592, "count": 1, "min": 279.0954113006592, "max": 279.0954113006592}, "epochs": {"sum": 15.0, "count": 1, "min": 15, "max": 15}, "check_early_stopping.time": {"sum": 2.2656917572021484, "count": 6, "min": 0.1614093780517578, "max": 0.8072853088378906}, "update.time": {"sum": 176755.446434821, "count": 5, "min": 34637.386322021484, "max": 36447.35622406006}, "finalize.time": {"sum": 3536.5140438079834, "count": 1, "min": 3536.5140438079834, "max": 3536.5140438079834}, "setup.time": {"sum": 2.808809280395508, "count": 1, "min": 2.808809280395508, "max": 2.808809280395508}, "total.time": {"sum": 181544.9457168579, "count": 1, "min": 181544.9457168579, "max": 181544.9457168579}}]}

2026-04-29 11:39:46 Uploading - Uploading generated training model
2026-04-29 11:39:46 Completed - Training job completed
Training seconds: 341
Billable seconds: 341

Perform a batch prediction by using the newly trained model.

[69]: # Enter your code here
test_labels, target_predicted = batch_linear_predict(test, classifier_estimator2)

INFO:sagemaker.image_uris:Same Images used for training and inference. Defaulting to image scope: inference.
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
INFO:sagemaker:Creating model with name: linear-learner-2026-04-29-11-40-13-535
INFO:sagemaker:Creating transform job with name: linear-learner-2026-04-29-11-40-14-138
.....Docker entrypoint called with argument(s): serve
Running default environment configuration script
[04/29/2026 11:47:04 INFO 140622025713472] Memory profiler is not enabled by the environment variable ENABLE_PROFILER.
/opt/amazon/lib/python3.8/site-packages/mxnet/model.py:97: SyntaxWarning: "is" with a literal. Did you mean "=="?
if num_device is 1 and 'dist' not in kvstore:
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:495: SyntaxWarning: "is" with a literal. Did you mean "=="?
if cons['type'] is 'ineq':
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:743: SyntaxWarning: "is not" with a literal. Did you mean "!="?
if len(self.X_min) is not 0:
[04/29/2026 11:47:08 WARNING 140622025713472] Loggers have already been setup.
[04/29/2026 11:47:08 INFO 140622025713472] loaded entry point class algorithm.serve.server_config:config_api
[04/29/2026 11:47:08 INFO 140622025713472] loading entry points
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/json
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/jsonlines
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator application/x-recordio-protobuf
[04/29/2026 11:47:08 INFO 140622025713472] loaded request iterator text/csv
[04/29/2026 11:47:08 INFO 140622025713472] loaded response encoder application/json
[04/29/2026 11:47:08 INFO 140622025713472] loaded response encoder application/jsonlines

```

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 2, Col 28 Flight_Delay-Student.ipynb

Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/en_us/Flight_Delay-Student.ipynb

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Flight_Delay-Student.ipynb

conda_python3

```

actions.count": {"sum": 1.0, "count": 1, "min": 1, "max": 1}}]
2026-04-29T11:47:09.195:[sagemaker logs]: MaxConcurrentTransforms=4, MaxPayloadInMB=6, BatchStrategy=MULTI_RECORD

Plot a confusion matrix.

[70]: # Enter your code here
plot_confusion_matrix(test_labels, target_predicted)

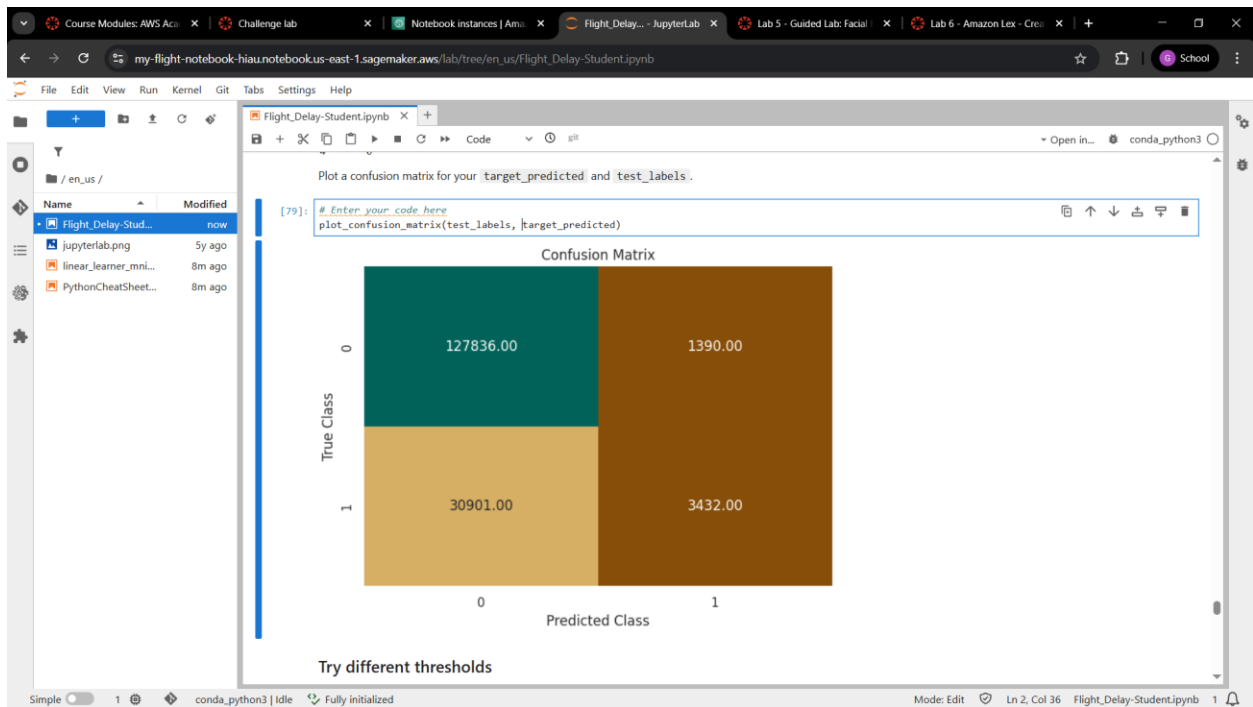
```

Confusion Matrix

	0	1
True Class 0	127657.00	1569.00
True Class 1	32302.00	2031.00

The linear model shows only a little improvement in performance. Try a tree-based ensemble model, which is called XGBoost, with Amazon SageMaker.

Simple 1 conda_python3 | Idle Fully initialized Mode: Edit Ln 2, Col 27 Flight_Delay-Student.ipynb



Course Modules: AWS Acc... Challenge lab Notebook instances | Ami... Flight_Delay... - JupyterLab Lab 5 - Guided Lab: Facial Lab 6 - Amazon Lex - Cre...

my-flight-notebook-hiau.notebook.us-east-1.sagemaker.aws/lab/tree/en_us/Flight_Delay-Student.ipynb

File Edit View Run Kernel Git Tabs Settings Help

Flight_Delay-Student.ipynb

Try different thresholds

Question: Based on how well the model handled the test set, what can you conclude?

#Enter your answer here

XGBoost outperforms the linear model in AUC and recall, and adjusting the threshold trades off precision against recall where the optimal value depends on whether missing a real delay or sending a false alarm is more costly to the business

Hyperparameter optimization (HPO)

```
[80]: from sagemaker.tuner import IntegerParameter, CategoricalParameter, ContinuousParameter, HyperparameterTuner

## You can spin up multiple instances to do hyperparameter optimization in parallel

xgb = sagemaker.estimator.Estimator(container,
                                     role=sagemaker.get_execution_role(),
                                     instance_count=1, # make sure you have a limit set for these instances
                                     instance_type=instance_type,
                                     output_path='s3://{}(/)/output'.format(bucket, prefix),
                                     sagemaker_session=sess)

xgb.set_hyperparameters({'eval_metric':'auc'}
```

Simple 1 conda_python3 | Idle Fully initialized Mode: Command Ln 3, Col 240 Flight_Delay-Student.ipynb 1

Conclusion

You have now iterated through training and evaluating your model at least a couple of times. It's time to wrap up this project and reflect on:

- What you learned
- What types of steps you might take moving forward (assuming that you had more time)

Use the following cell to answer some of these questions and other relevant questions:

- Does your model performance meet your business goal? If not, what are some things you'd like to do differently if you had more time for tuning?
- How much did your model improve as you made changes to your dataset, features, and hyperparameters? What types of techniques did you employ throughout this project, and which yielded the greatest improvements in your model?
- What were some of the biggest challenges that you encountered throughout this project?
- Do you have any unanswered questions about aspects of the pipeline that didn't make sense to you?
- What were the three most important things that you learned about machine learning while working on this project?

Project presentation: Make sure that you also summarize your answers to these questions in your project presentation. Combine all your notes for your project presentation and prepare to present your findings to the class.

#Write your answers here

- The XGBoost model with HPO approaches the business goal but recall on delays could still be improved with SMOTE resampling and richer real-time features
- The biggest improvement was to switch to XGBoost, with additional improvements from adding weather and holiday features, and incremental gains from HPO
- Class imbalance skewed accuracy readings, weather imputation required extra handling, and SageMaker training were slow takes hours to train
- None
- Proper data is needed for good results as poor quality or imbalanced data directly limits model performance regardless of algorithm choice, ensemble models like XGBoost consistently outperform linear models on structured tabular data

Canvas Details

Credentials

Cloud Access

Cloud Labs

Remaining session time: 00:18:44 (19 minutes)
 Session started at: 2026-04-29T01:11:58-0700
 Session to end at: 2026-04-29T07:11:58-0700

Accumulated lab time: 05:40:00 (340 minutes)

No running instance

SSH key: [Show](#) [Download PEM](#) [Download PPK](#)

AWS SSO: [Download URL](#)

JOSE RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES AND ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

CPE C313

ARTIFICIAL INTELLIGENCE

LABORATORY

6

Pozon, Gion Terrence B.

301G

2026-04-29

Lab 6: Amazon Lex

AWS Console account

The screenshot shows the Amazon Lex console interface. The left sidebar contains navigation links for Bots, Bot templates, Networks of bots, Test workbench, Test sets, Test results, and Related resources. The main content area displays a notification about Amazon Lex Generative AI features powered by Amazon Bedrock. Below the notification, the 'Bots (0)' section shows a search bar and a table with columns: Name, Description, Status, Latest Version, and Last updated. A 'Create bot' button is visible. The 'Import/export history (0)' section also shows a search bar and a 'Delete selected' button.

Task 3

The screenshot shows the Amazon Lex console interface after creating a bot. The 'Bots (1)' section now displays a table with one bot:

Name	Description	Status	Latest Version	Last updated
ScheduleAppointment	-	Available	-	6 minutes ago

The 'Import/export history (0)' section remains empty. The URL bar at the bottom shows the path: `https://us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bots`.

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 x +

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment**
- Bot versions
 - Draft version
 - All languages
 - English (US)
 - Intents
 - Slot types
- Deployment
 - Aliases
- Channel integrations
- Analytics [New](#)
- Overview
- Conversation dashboard
- Conversation flows
- Conversations
- Performance dashboard

ScheduleAppointment

Lex > Bots > Bot: ScheduleAppointment

Bot details [Edit](#)

Name
ScheduleAppointment

Description
-

ID: RVNVWDHR8F

Add languages [Info](#)



Add languages so your bot can engage in conversations. Each language for your bot should be configured with intents to accomplish a goal, sample utterances allow to start a dialog, slots to collect additional information, and fulfillment to complete user's request.

[View languages](#) 1 language

Create versions and aliases for deployment [Info](#)

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 x +

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/aliases

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment**
- Bot versions
 - Draft version
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 - Slot types
- Deployment
 - Aliases**
- Channel integrations
- Analytics [New](#)
- Overview
- Conversation dashboard
- Conversation flows
- Conversations
- Performance dashboard

Aliases (1) [Info](#)

An alias points to a specific version of your bot. With an alias, you can update the bot version that your client applications use.

[Delete](#) [Create alias](#)

Alias name	Created	Associated version
☐ TestBotAlias	7 minutes ago	Draft version

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment
 - Bot versions
 - Draft version
 - All languages
 - English (US)
 - Intents
 - Slot types
 - Deployment
 - Aliases**
 - Channel integrations
 - Analytics [New](#)
 - Overview
 - Conversation dashboard
 - Conversation flows
 - Conversations
- Performance dashboard

Alias: TestBotAlias [Info](#)

[Delete](#) [Associate version with alias](#)

Info This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.

Details

[Edit](#)

Alias name	Description
TestBotAlias	test bot alias
Associated version	Sentiment analysis-
Draft version	Disabled
ID: TSTALIASID	

Languages (1) [Info](#)

[Test](#) [Manage languages in alias](#)

Select the languages you want to enable in the alias. Only languages that are built can be enabled.

< 1 > [Refresh](#)

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID/locale/en_US/

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID/locale/en_US/

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment
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Alias language support: English (US)

Lambda function - optional
This Lambda function is invoked for initialization, validation, and fulfillment.

Source:

Lambda function version or alias:

[Learn more about Lambda](#)

[Cancel](#) [Save](#)

Course Modules: All

Challenge lab

AWS Management C

my-flight-notebook

Lab 5 - Guided Lab

Lab 6 - Amazon Lex

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/alias/TSTALIASID

us

Search

[Alt+S]

United States (N. Virginia)

Account ID: 1192-8573-5731

vociabs/user5004810=POZON,TERRENCE

Amazon Lex

Bots

Bot templates New

Networks of bots New

ScheduleAppointment

Bot versions

Draft version

All languages

English (US)

Intents

Slot types

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Analytics New

Overview

Conversation dashboard

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Conversations

Performance dashboard

Language settings updated successfully

Lex > Bots > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias

Alias: TestBotAlias Info

Delete

Associate version with alias

This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.

Details

Edit

Alias name

TestBotAlias

Description

test bot alias

Associated version

Draft version

Sentiment analysis

Disabled

ID: TSTALIASID

Languages (1) Info

Test

Manage languages in alias

Select the languages you want to enable in the alias. Only languages that are built can be enabled.

CloudShell

Feedback

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Task 4

The image displays two screenshots of the Amazon Lex console, showing the 'Intents' page for a bot named 'ScheduleAppointment'.

Top Screenshot: The 'Intents' page is shown with the 'Draft version' selected. The 'Intents' list contains two items:

Name	Description	Last edited
MakeAppointment	Intent to book a dentist's appointment	8 minutes ago
FallbackIntent	Default intent when no other intent matches	9 minutes ago

Bottom Screenshot: The 'Intents' page is shown with the 'Building' status. A blue banner at the top reads: "Building language English (US) in bot: ScheduleAppointment. If your language contains external source slot types, the build might take longer to complete." The 'Intents' list remains the same as in the top screenshot.

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment
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 - All languages
 - English (US)**
 - Intents**
 - Slot types
 - Deployment
 - Aliases
 - Channel integrations
 - Analytics [New](#)
 - Overview
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 - Performance dashboard

Successfully built language English (US) in bot: ScheduleAppointment

Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents

Draft version English (US) Successfully built Analyze Build Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
☐	MakeAppointment	Intent to book a dentist's appointment
☐	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: Now

Inspect

Ready for complete testing

Type a message

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment
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 - Intents**
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 - Analytics [New](#)
 - Overview
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 - Conversation flows
 - Conversations
 - Performance dashboard

Successfully built language English (US) in bot: ScheduleAppointment

Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents

Draft version English (US) Successfully built Analyze Build Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
☐	MakeAppointment	Intent to book a dentist's appointment
☐	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

I would like to make an appointment

What type of appointment would you like to schedule?

A root canal

When would you like to schedule your root canal?

tomorrow

Ready for complete testing

Type a message

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 x +

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- ScheduleAppointment
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 - Intents**
 - Slot types
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 - Aliases
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 - Analytics [New](#)
 - Overview
 - Conversation dashboard
 - Conversation flows
 - Conversations
 - Performance dashboard

Successfully built language English (US) in bot: ScheduleAppointment

Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents

Draft version English (US) Successfully built Analyze Build Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

Name	Description
☐ MakeAppointment	Intent to book a dentist's appointment
☐ FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

A root canal

When would you like to schedule your root canal?

tomorrow

What time on 2026-04-30 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.

10:00 a.m.

Ready for complete testing

Type a message

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Amazon Lex | us-east-1 x +

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

- Bots
- Bot templates [New](#)
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 - Analytics [New](#)
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 - Conversation flows
 - Conversations
 - Performance dashboard

Successfully built language English (US) in bot: ScheduleAppointment

Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents

Draft version English (US) Successfully built Analyze Build Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

Name	Description
☐ MakeAppointment	Intent to book a dentist's appointment
☐ FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

What time on 2026-04-30 works for you? We have plenty of availability, including 10:00 a.m., 10:30 a.m. and 11:00 a.m.

10:00 a.m.

10:00 is available, should I go ahead and book your appointment?

Yes

Ready for complete testing

Type a message

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Course Modules: All

Challenge lab

AWS Management C

my flight notebook

Lab 5 - Guided Lab

Lab 6 - Amazon Lex

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDHR8F/locale/en_US/intents

Search [Alt+S]

United States (N. Virginia)

Account ID: 1192-8573-5731
voclabs/user5004810=POZON_TERRENCE

Amazon Lex

Bots

Bot templates New

Networks of bots New

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English (US)

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Performance dashboard

Successfully built language English (US) in bot: ScheduleAppointment

Lex > Bots > Bot: ScheduleAp... > Versions > Version: Draft > All languages > Language: English (US) > Intents

Draft version

English (US)

Successfully built

Analyze

Build

Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
☐	MakeAppointment	Intent to book a dentist's appointment
☐	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

10:00 is available, should I go ahead and book your appointment?

Yes

Intent MakeAppointment is fulfilled

Ready for complete testing

Type a message

CloudShell

Feedback

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Task 5

The screenshot shows the AWS Management Console for Amazon Cognito Identity Pools. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/cognito/v2/identity/identity-pools?region=us-east-1`. The console header shows the user is logged in as `voclabs/user5004810=POZON_TERRENCE` with account ID `1192-8573-5731`.

Identity pools (1) Info

View and configure your identity pools. Identity pools enable you to create unique identities and assign permissions for users.

Search identity pools by name or ID

Identity pool name	Identity pool ID	Created time	Last updated time
MyIdentityPool	us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f	21 minutes ago	20 minutes ago

MyIdentityPool Info

[Rename](#) [Delete identity pool](#)

Identity pool overview

Identity pool name MyIdentityPool	ARN arn:aws:cognito-identity:us-east-1:119285735731:identitypool/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f	Created time April 29, 2026 at 22:14 GMT+8
Identity pool ID us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f	User access Guest	Last updated time April 29, 2026 at 22:15 GMT+8

Get started with your Amazon Cognito identity pool

[User statistics](#) | [Identity browser](#) | [User access](#) | [Identity pool properties](#) | [Other properties](#)

Identity overview

Identities this month 0	Total identities 0	Unauthenticated 0%	Authenticated 0%
-----------------------------------	------------------------------	------------------------------	----------------------------

Identities

CloudShell Feedback

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Course Modules: All

Challenge lab

AWS Management C

my-flight-notebook

Lab 5 - Guided Lab

Lab 6 - Amazon Lex

User statistics > My

us-east-1.console.aws.amazon.com/cognito/v2/identity/identity-pools/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f/user-statistics?region=us-east-1

Search [Alt+S]

United States (N. Virginia)

Account ID: 1192-8573-5731
vociabs/user5004810=POZON, TERRENCE

Amazon Cognito

Identity pools

MyIdentityPool

RenameDelete identity pool

User pools

Identity pools

Identity pool overview

Identity pool name

ARN

Created time

Last updated time

User access

us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f

arn:aws:cognito-identity:us-east-1:119285735731:identitypool/us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f

April 29, 2026 at 22:14 GMT+8

April 29, 2026 at 22:15 GMT+8

Guest

Get started with your Amazon Cognito identity pool

User statistics

Identity browser

User access

Identity pool properties

Other properties

Identity overview

Identities this month

Total identities

Unauthenticated

Authenticated

0

0

0%

0%

Identities

CloudShell

Feedback

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Task 6

The screenshot displays the AWS Academy interface for Task 6: Creating an S3 bucket. The task instructions are as follows:

Now that you set up the security permissions, you must create an S3 bucket to host your webpage.

30. First, download the following two webpage files and extract them to a local file directory.

- index.html
- error.html

The index.html file includes the script that will load your bot.

Download the .zip file.

31. On the AWS Management Console, on the **Services** menu, choose **S3**.

32. On the **Amazon S3** page, choose **Create bucket**.

33. Enter a name for your bucket. Replace all

The file explorer window shows the following files:

Name	Date modified	Type	Size
cleaned_dataset\			
New folder			
New folder (2)			
New folder (2)			
error.html	3 Apr 2024 3:53 pm	Vivaldi HTML Doc...	3 KB
index.html	19 May 2025 12:15 am	Vivaldi HTML Doc...	10 KB
lab6.zip	29 Apr 2026 10:37 pm	Compressed (zip...	4 KB

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Homepage | S3 | us x +

us-east-1.console.aws.amazon.com/s3/get-started?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Storage

Amazon S3

Store and retrieve any amount of data from anywhere

Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

Create bucket

Pricing

With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.

Estimate your monthly bill using the [AWS Simple Monthly Calculator](#)

[View pricing details](#)

Resources

[User guides](#)

How it works

Introduction to Amazon S3 | Amazon Web Services

Amazon Web Services



Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Create S3 bucket | S3 x +

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

amzn-s3-demo-bucket

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Create S3 bucket [5] x +

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > Create bucket

By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

lexlab6

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Create S3 bucket [5] x +

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > Create bucket

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you

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Amazon S3 > Buckets > lexlab6

Successfully created bucket "lexlab6"

lexlab6 Info

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (0)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name Type Last modified Size Storage class

No objects

You don't have any objects in this bucket.

Upload

Amazon S3 > Buckets > lexlab6 > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (0)

All files and folders in this table will be uploaded.

Find by name

Name Folder Type Size

No files or folders

You have not chosen any files or folders to upload.

Destination Info

Destination

s3://lexlab6

Destination details

Bucket settings that impact new objects stored in the specified destination.

Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: l x Lab 6 - Amazon Lex x Upload objects - S3 x

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1

Search [Alt+S] Ask Amazon Q

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Upload

Upload Info

Add the files and folders you want to upload to S3.

Files and folders (0)

All files and folders in this bucket will be uploaded.

Find by name

Name

Destination Info

Destination

s3://lexlab6

► Destination details

Bucket settings that impact new objects stored in the specified destination.

Open

File name: "index.html" "error.html"

All Files (*.*)

Open Cancel

Add folder.

Remove Add files Add folder

< 1 >

Size

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: l x Lab 6 - Amazon Lex x Upload objects - S3 x

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1

Search [Alt+S] Ask Amazon Q

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (2 total, 12.6 KB)

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
☐	error.html	-	text/html	2.9 KB
☐	index.html	-	text/html	9.7 KB

Destination Info

Destination

s3://lexlab6

► Destination details

Bucket settings that impact new objects stored in the specified destination.

► Permissions

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Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / Upload objects - S3

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3

Upload succeeded
For more information, see the Files and folders table.

Upload: status
After you navigate away from this page, the following information is no longer available.

Summary
Destination: s3://lexlab6
Succeeded: 2 files, 12.6 KB (100.00%)
Failed: 0 files, 0 B (0%)

Files and folders Configuration

Files and folders (2 total, 12.6 KB)
Find by name

Name	Folder	Type	Size	Status	Error
error.html	-	text/html	2.9 KB	Succeeded	-
index.html	-	text/html	9.7 KB	Succeeded	-

Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / lexlab6 - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=objects

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6

lexlab6 Info
Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (2)
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)
Find objects by prefix

Name	Type	Last modified	Size	Storage class
error.html	html	April 29, 2026, 22:42:09 (UTC+08:00)	2.9 KB	Standard
index.html	html	April 29, 2026, 22:42:10 (UTC+08:00)	9.7 KB	Standard

Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / lexlab6 - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6

lexlab6 Info

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Bucket overview

AWS Region
US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)
arn:aws:s3:::lexlab6

Creation date
April 29, 2026, 22:40:01 (UTC+08:00)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Disabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Bucket ABAC

Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. [Learn more](#)

ABAC status

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Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / Edit static website h

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6 Edit static website hosting

Edit static website hosting Info

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☒ Disable

☐ Enable

Cancel Save changes

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Course Modules: All | Challenge lab | AWS Management C | my-flight-notebook | Lab 5 - Guided Lab | Lab 6 - Amazon Lex | Edit static website h | +

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Edit static website hosting

Edit static website hosting [info](#)

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional

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Course Modules: All | Challenge lab | AWS Management C | my-flight-notebook | Lab 5 - Guided Lab | Lab 6 - Amazon Lex | Edit static website h | +

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/website/edit?region=us-east-1

Search [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Edit static website hosting

☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

1	
---	--

Course Modules: All / Challenge lab AWS Management C my-flight-notebook Lab 5 - Guided Lab: Lab 6 - Amazon Lex lexlab6 - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties

us-east-1 console.aws.amazon.com

Search [Alt+S] Ask Amazon Q

United States (N. Virginia)

Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6

Successfully edited static website hosting.

Tags

You can use tags to track storage costs, organize general purpose buckets, and specify permissions for a general purpose bucket. AWS-generated tags are created by AWS and are read-only. [Learn more](#)

S3 Console now uses s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets by default. To use these APIs for tagging, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. [Learn more](#)

User-defined tags AWS-generated tags

User-defined tags (0) Info

Refresh Delete Add new tag

Key Value - optional

There are no user-defined tags associated with this bucket.

Add new tag

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Edit

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Task 7

The screenshot displays a web browser window with two main sections. The top section is a code editor showing the HTML and JavaScript code for a chatbot interface. The bottom section is the Amazon Lex console, specifically the 'ScheduleAppointment' bot details page.

Code Editor (index.html):

```
<html>
<head>
</head>
<body>
<h1 style="text-align: left">Amazon Lex - Appointment BOT</h1>
<div id="searchResults" style="width: 380px; height: 450px; border: 1px solid #ccc; background-color: #eee; padding: 4px; overflow: scroll; margin-right: 10px; float: left; display:
<div id="conversation" style="width: 380px; height: 450px; border: 1px solid #ccc; background-color: #eee; padding: 4px; overflow: scroll"></div>
<form id="chatform" style="margin-bottom: 10px" onsubmit="return pushChat();">
  <p style="width: 380px; font-size: 0.8em; line-height: 1.2em">
    <input type="text" id="wisdom" size="80" value="" placeholder="What do you want to do?">
  </p>
</form>
<script type="text/javascript">
  // set the focus to the input box
  document.getElementById("wisdom").focus();
  // Initialize the Amazon Cognito credentials provider
  //replace the
  AWS.config.region = 'us-east-1'; // Region
  AWS.config.credentials = new AWS.CognitoIdentityCredentials({
    IdentityPoolId: 'us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f',
  });
  var lexRuntimeV2 = new AWS.LexRuntimeV2();
  var lexUserId = 'chatbot-demo' + Date.now();
  var sessionAttributes = {};
  // Configuration for your Lex V2 bot
  const BOT_ID = 'SA0Y5M4SV8'; // Your Bot ID
  const BOT_ALIAS_ID = 'TSTALIASID'; // Your Bot Alias ID
  const LOCALE_ID = 'en-US';
  function pushChat() {
    var wisdomText = document.getElementById('wisdom');
    var wisdom = wisdomText.value.trim();
  }
```

Amazon Lex Console:

- Bot details:**
 - Name: ScheduleAppointment
 - Description: -
 - ID: RVNVWDHR8F
- Add languages:**
 - 1 language (English (US))
- Create versions and aliases for deployment:**

index.html

C:\Users> OhH! > Downloads > lab6 > index.html > html > body > script > BOT_ID

2 <html>

127 <body>

137

138 <script type="text/javascript">

139 // set the focus to the input box

140 document.getElementById("wisdom").focus();

141

142 // Initialize the Amazon Cognito credentials provider

143 //replace the

144 AWS.config.region = 'us-east-1'; // Region

145 AWS.config.credentials = new AWS.CognitoIdentityCredentials({

146 IdentityPoolId: 'us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f',

147 });

148

149 var lexRuntimeV2 = new AWS.LexRuntimeV2();

150 var lexUserId = 'chatbot-demo' + Date.now();

151 var sessionAttributes = {};

152

153 // Configuration for your Lex V2 bot

154 const BOT_ID = 'RVNVWDH8B'; // Your Bot ID

155 const BOT_ALIAS_ID = 'TSTALIASID'; // Your Bot Alias ID

156 const LOCALE_ID = 'en-US';

157

158 function pushChat() {

159 var wisdomText = document.getElementById('wisdom');

160 var wisdom = wisdomText.value.trim();

161

162 if (wisdom.length > 0) {

163 console.log('User input:', wisdom);

164

165 // disable input while processing

166 wisdomText.value = '';

167 wisdomText.locked = true;

168

169 // Show user's message

170 showRequest(wisdom);

171

172 // Prepare parameters for Lex V2

173

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/RVNVWDH8B/alias/TSTALIASID

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon Lex

Bots

Bot templates New

Networks of bots New

ScheduleAppointment

Bot versions

Draft version

All languages

English (US)

Intents

Slot types

Deployment

Aliases

Channel integrations

Analytics New

Overview

Conversation dashboard

Conversation flows

Conversations

Performance dashboard

Lex > Bots > Bot: ScheduleAppointment > Aliases > Alias: TestBotAlias

Alias: TestBotAlias Info

Delete Associate version with alias

This alias is intended only for testing. Use it to test parameters such as voice, session timeouts, and fulfillment logic. You can send a maximum of 2 requests per second to this alias. It is associated with the draft version by default. This association cannot be modified. You can't delete the test alias or deploy it on a channel. All languages in the draft version are included in the alias.

Details Edit

Alias name

TestBotAlias

Description

test bot alias

Associated version

Draft version

Sentiment analysis-

Disabled

ID: TSTALIASID

Languages (1) Info Test Manage languages in alias

Select the languages you want to enable in the alias. Only languages that are built can be enabled.

Search languages

1

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File Edit Selection View Go Run Terminal Help

Search

Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More

index.html

C:\Users> OhH! > Downloads > lab6 > index.html > html > body > script > BOT_ALIAS_ID

2 <html>

127 <body>

137

138 <script type="text/javascript">

139 // set the focus to the input box

140 document.getElementById("wisdom").focus();

141

142 // Initialize the Amazon Cognito credentials provider

143 //replace the

144 AWS.config.region = 'us-east-1'; // Region

145 AWS.config.credentials = new AWS.CognitoIdentityCredentials({

146 IdentityPoolId: 'us-east-1:7e7d9d71-41f6-4f97-844a-5516aab8e56f',

147 });

148

149 var lexRuntimeV2 = new AWS.LexRuntimeV2();

150 var lexUserId = 'chatbot-demo' + Date.now();

151 var sessionAttributes = {};

152

153 // Configuration for your Lex V2 bot

154 const BOT_ID = 'RVMMQHR8R'; // Your Bot ID

155 const BOT_ALIAS_ID = 'ISTALIASID'; // Your Bot Alias ID

156 const LOCALE_ID = 'en-us';

157

158 function pushChat() {

159 var wisdomText = document.getElementById('wisdom');

160 var wisdom = wisdomText.value.trim();

161

162 if (wisdom.length > 0) {

163 console.log('User input:', wisdom);

164

165 // disable input while processing

166 wisdomText.value = '';

167 wisdomText.locked = true;

168

169 // Show user's message

170 showRequest(wisdom);

171

172 // Prepare parameters for Lex V2

173

Ln 155, Col 33 (10 selected) Spaces: 4 UTF-8 (LF) HTML

Course Modules: All Challenge lab AWS Management C my-flight-notebook Lab 5 - Guided Lab Lab 6 - Amazon Lex lexlabs - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=objects

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlabs

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization settings

lexlab6

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (2)

Copy S3 URI Copy URL Download Open i Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
	error.html	html	April 29, 2026, 22:42:09 (UTC+08:00)	2.9 KB	Standard
	index.html	html	April 29, 2026, 22:42:10 (UTC+08:00)	9.7 KB	Standard

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Upload objects - S3 x

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)
All files and folders in this table will be uploaded.

NameFolderTypeSize

No files or folders
You have not chosen any files or folders to upload.

Destination Info
Destination
[s3://lexlab6](#)
► Destination details
Bucket settings that impact new objects stored in the specified destination.

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Course Modules: All x Challenge lab x AWS Management C x my-flight-notebook x Lab 5 - Guided Lab: x Lab 6 - Amazon Lex x Upload objects - S3 x

us-east-1.console.aws.amazon.com/s3/upload/lexlab6?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

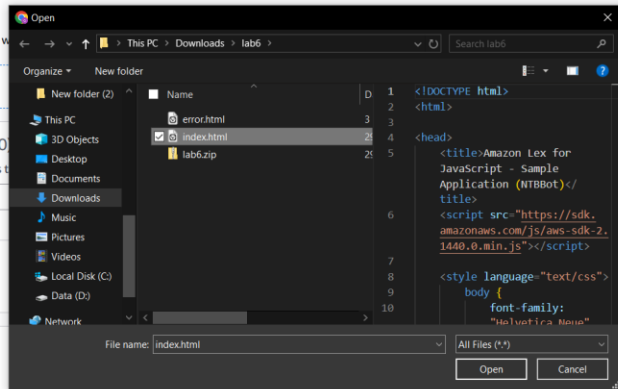
Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)
All files and folders in this table will be uploaded.

NameFolderTypeSize

No files or folders
You have not chosen any files or folders to upload.

Destination Info
Destination
[s3://lexlab6](#)
► Destination details
Bucket settings that impact new objects stored in the specified destination.



Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://lexlab6

Succeeded
1 file, 9.7 KB (100.00%)

Failed
0 files, 0 B (0%)

[Files and folders](#) Configuration

Files and folders (1 total, 9.7 KB)

Find by name							< 1 >	
Name	Folder	Type	Size	Status	Error			
index.html	-	text/html	9.7 KB	Succeeded	-			

Amazon S3 > Buckets > lexlab6

lexlab6 Info

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).
[View analyzer for us-east-1](#)

Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access

Off

► Individual Block Public Access settings for this bucket

Bucket policy

Edit

Delete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Course Modules: AllChallenge labAWS Management Cmy-flight-notebookLab 5 - Guided LabLab 6 - Amazon LexEdit bucket policy

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1

Search[Alt+S]United States (N. Virginia)Account ID: 1192-8573-5731voclabs/user5004810=POZON_TERRENCE

Amazon S3Bucketslexlab6Edit bucket policy

Edit bucket policy Info

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

arn:aws:s3:::lexlab6

Policy

1

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

Course Modules: AllChallenge labAWS Management Cmy-flight-notebookLab 5 - Guided LabLab 6 - Amazon LexEdit bucket policy

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1

Search[Alt+S]United States (N. Virginia)Account ID: 1192-8573-5731voclabs/user5004810=POZON_TERRENCE

Amazon S3Bucketslexlab6Edit bucket policy

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3:::example.com/*"
10    },
11    {
12      "Sid": "AllowLexV2Runtime",
13      "Effect": "Allow",
14      "Principal": {
15        "Service": "lexv2.amazonaws.com"
16      },
17      "Action": [
18        "s3:GetObject",
19        "s3:PutObject"
20      ],
21      "Resource": "arn:aws:s3:::example.com/*"
22    }
23  ]
24 }
25
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

Course Modules: All | Challenge lab | AWS Management C | my-flight-notebook | Lab 5 - Guided Lab | Lab 6 - Amazon Lex | Edit bucket policy -

us-east-1.console.aws.amazon.com/s3/bucket/lexlab6/property/policy/edit?region=us-east-1

Search [Alt+S] United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6 > Edit bucket policy

```
5 {
6   "Effect": "Allow",
7   "Principal": "*",
8   "Action": "s3:GetObject",
9   "Resource": "arn:aws:s3:::lexlab6/*"
10 },
11 {
12   "Sid": "AllowLexV2Runtime",
13   "Effect": "Allow",
14   "Principal": {
15     "Service": "lexv2.amazonaws.com"
16   },
17   "Action": [
18     "s3:GetObject",
19     "s3:PutObject"
20   ],
21   "Resource": "arn:aws:s3:::lexlab6/*"
22 }
23 ]
24 }
25 }
```

+ Add new statement

JSON Ln 9, Col 38

Security: 1 Errors: 0 Warnings: 0 Suggestions: 0

Learn more about policy validation

Choose a service

Search services

Included

S3

Available

AI Operations

AMP

API Gateway

API Gateway V2

ARC Region switch

ARC Zonal Shift

ASC

Add a resource Add

Add a condition (optional) Add

Preview external access

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Course Modules: All | Challenge lab | AWS Management C | my-flight-notebook | Lab 5 - Guided Lab | Lab 6 - Amazon Lex | lexlab6 - S3 bucket |

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=permissions

Search [Alt+S] United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 > Buckets > lexlab6

Successfully edited bucket policy.

```
{
  "Sid": "PublicReadGetObject",
  "Effect": "Allow",
  "Principal": "*",
  "Action": "s3:GetObject",
  "Resource": "arn:aws:s3:::lexlab6/*"
},
{
  "Sid": "AllowLexV2Runtime",
  "Effect": "Allow",
  "Principal": {
    "Service": "lexv2.amazonaws.com"
  },
  "Action": [
    "s3:GetObject",
    "s3:PutObject"
  ],
  "Resource": "arn:aws:s3:::lexlab6/*"
}
]
```

Object Ownership Edit

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

Bucket owner enforced

ACLs are disabled. All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

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Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / lexlab6 - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties

Search [Alt+S] United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6

lexlab6 Info

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Bucket overview

AWS Region
US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)
arn:aws:s3::lexlab6

Creation date
April 29, 2026, 22:40:01 (UTC+08:00)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Disabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Bucket ABAC

Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. [Learn more](#)

ABAC status

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Course Modules: All / Challenge lab / AWS Management C / my-flight-notebook / Lab 5 - Guided Lab / Lab 6 - Amazon Lex / lexlab6 - S3 bucket

us-east-1.console.aws.amazon.com/s3/buckets/lexlab6?region=us-east-1&tab=properties

Search [Alt+S] United States (N. Virginia) Account ID: 1192-8573-5731 voclabs/user5004810=POZON_TERRENCE

Amazon S3 Buckets lexlab6

Requester pays

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting
Enabled

Hosting type
Bucket hosting

Bucket website endpoint copied

When you configure your static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://lexlab6-s3-website-us-east-1.amazonaws.com>

CloudShell Feedback

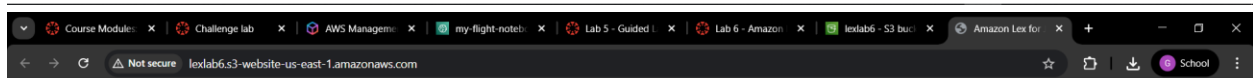
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Amazon Lex - Appointment BOT

Hi there! What do you want to do?

What do you want to do?



Amazon Lex - Appointment BOT

Hi there! What do you want to do?

can you make me an appointment?

What type of appointment would you like to schedule?

what are available?

What type of appointment would you like to schedule?

a dining reservation

What type of appointment would you like to schedule?

root canal

When would you like to schedule your root canal?

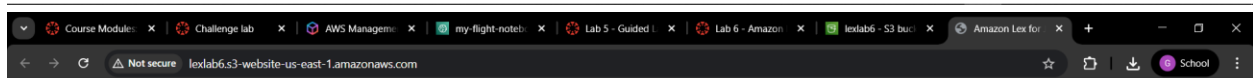
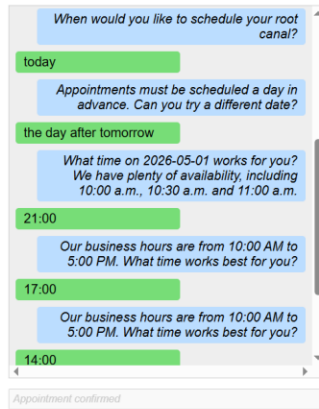
today

Appointments must be scheduled a day in advance.

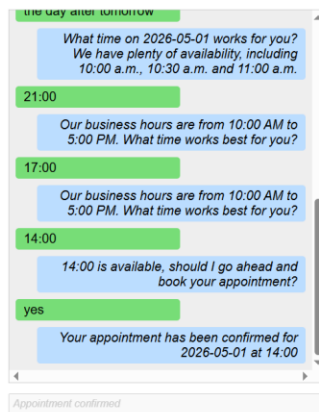
Appointment confirmed



Amazon Lex - Appointment BOT



Amazon Lex - Appointment BOT



Canvas Details

The screenshot displays the AWS Academy web application. The browser's address bar shows the URL `awsacademy.instructure.com/courses/168322/modules/items/16494129`. The page header indicates the current location: `AMLFv1EN-LTI13-168322 > Modules > Module 6 - Introducing Natural Language Processing > Lab 6 - Amazon Lex - Create a chatbot`.

On the left, a sidebar contains navigation links: Home, Account, Modules, Discussions, Grades (with a notification badge), Courses, Lucid (Whiteboard), Calendar, Inbox, History, and Help (with a notification badge).

The main content area features a 'Credentials' modal window. This window is titled 'Credentials' and has a close button in the top right corner. It is divided into two sections: 'Cloud Access' and 'Cloud Labs'.

The 'Cloud Access' section includes:

- AWS CLI:** A 'Show' button.
- Cloud Labs:** A section containing session details:
 - Remaining session time: 02:14:13 (135 minutes)
 - Session started at: 2026-04-29T07:14:56-0700
 - Session to end at: 2026-04-29T10:12:56-0700
 - Accumulated lab time: 00:43:00 (43 minutes)
 - No running instance
- SSH key:** 'Show', 'Download PEM', and 'Download PPK' buttons.
- AWS SSO:** 'Download URL' button.

The 'Cloud Labs' section also includes a 'Source' checkbox and a 'Back' button.

At the bottom of the modal, a message states: 'You should see the following confirmation.' Below the modal, there are 'Previous' and 'Next' navigation buttons.